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Techno-economic analysis of the lithium-ion and lead-acid battery in microgrid systems

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ABSTRACT

Microgrids are a beneficial alternative to the conventional generation system that can provide greener, reliable and high quality power with reduced losses, and lower network congestion. However, the performance of renewable energy resource (RER) based generators in a microgrid is hindered by their intermittent nature. The energy storage system plays a key role in overcoming the intermittency of renewable sources by balancing the power demand against variable generation. Energy storage using batteries is accepted as one of the most important and efficient ways for retaining reliable energy supply whilst incorporating RERs into the electricity grid. Lead-acid (LA) batteries have been the most commonly used electrochemical energy storage technology for grid-based applications till date, but many other competing technologies are also being used such as lithium-ion (Li-ion), Sodium-Sulphur and flow batteries. This paper carries out the techno-economic analysis of the battery storage system under different configurations of the microgrid system. The design of an optimal model of standalone as well as grid-connected microgrid systems having PV-wind-diesel and biodiesel energy resources in the presence of Li-ion ($LiFeSO_4$ type) and LA batteries have been studied. The performance of the two types of batteries has been compared in the proposed microgrid systems considering realistic load profiles, real resource data, and real prices of the components. Load profiles of two rural villages, a business organization and an urban residential building have been used. On the basis of these load profiles and available resources, different microgrid configurations with generation from PV, wind turbine, diesel and biodiesel generators have been considered to study the performances of the batteries. The results include the cost of optimal system configuration under the given scenarios, electricity generation by various generating sources, detail comparison between various components, emissions, and various batteries performance assessment parameters. The study reveals that Li-ion batteries as energy storage are techno-economically more viable compared to LA batteries and they are expected to play a significant role in various applications of future electric power systems.

1. Introduction

Renewable energy in the electricity sector cannot only help in meeting the globally growing energy demand, but also can support the transformation of the existing grid into a smart grid. Wind and solar photovoltaic (PV) based renewable energy systems (RESs) have had substantial growth in the past decade above other renewable energy technologies. They can make electricity available to isolated and energy deficient regions [1]. Intermittent nature and increased levels of penetration of PV and wind power pose greater technical challenges to maintain the supply and demand balance of electricity in an electric power system [2]. These challenges motivate researchers across the globe to investigate and develop reliable energy storage systems (ESSs) that can offset this intermittency by storing the generated energy and making it available during times of demand [3]. In the last decade, the role of ESSs has increased in a variety of power system applications, such as grid stability enrichment, energy system efficiency enhancement, voltage-frequency regulation services, virtual inertia support, and in reducing the environmental impact from fossil fuel use [4,5].

Electrochemical energy storage technologies such as batteries are recognised as one of the most effective means of stabilising electrical networks with high levels of variable renewable energy (VRE). A range of battery chemistries can be used for energy storage in power system applications including load following, regulation, and energy management by adding or absorbing power from the grid [6]. Among different batteries, lead-acid (LA) type are the most commonly used ESS for electric power system applications. These batteries are well recognized for both automotive and industrial applications, and have been

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effectively implemented for utility storage [7]. The flooded lead–acid battery is a 150-year-old, matured and economical energy storage device, but has a short lifespan. This battery generally needs replacement every 4–5 years, which constitutes a major fraction of the system lifetime cost. Also, the future cost reductions of LA battery in commercial application can be anticipated to be small [8]. However, there are now rival electrochemical storage technologies including lithium-ion, sodium-sulphur and flow batteries that are effectively employed in energy storage applications.

Nowadays Lithium-ion (Li-ion) batteries are being used more in the power systems applications due to their lower maintenance, superior safety, volumetric and gravimetric energy densities characteristics. These features allow them to have less weight and size than that of LA batteries. In addition, the price of Li-ion battery technology is declining at 8–16% annually, and the cost advantage on Li-ion batteries is anticipated to improve considerably in the next few years [8].

Lithium-ion battery technology was developed commercially in the early 90s and it has empowered the portable electronics revolution. Currently this technology is increasingly being used in electric vehicles and grid storage applications [8]. The various Li-ion battery variants are further classified on the basis of employed chemistries such as energy density, cell voltage, cycle life, cost and rate capability (peukert factor), etc. The seven renowned Li-ion battery chemistries are LMO (lithium manganese oxide), NCA (lithium nickel-cobalt aluminium oxide), NMC (lithium nickel-manganese-cobalt oxide), LFP (lithium iron phosphate), LMO/LTO (lithium manganese-oxide cathode), NMC/LTO (lithium nickel-manganese-cobalt oxide cathode) and LFP/LTO (lithium iron-phosphate cathode). LFP/LTO includes the safety features of both LFP and LTO that make it safest among the seven chemistries. This chemistry has long cycle life with minimal degradation rates and has good performance under extreme temperature conditions. However, low cell voltage and high cost of LTO results in low energy density and high cost for the chemistry [8].

Because of the above discussed features and merits of Li-ion battery over LA battery, it is significant to compare the performances of both batteries as an energy storage unit in microgrid systems.

1.1. Literature review

Hybrid power systems based on renewable energy resources (RERs) can comprise a cost-effective substitute to fulfil the growing power demand. Because of the importance and benefits of HRES (Hybrid Renewable Energy Sources), numerous hybrid energy optimizations studies have been performed through Hybrid Optimization Model for Multiple Energy Resources (HOMER) and other software in the literature [9–31]. A detailed survey of various RES configurations, battery types, cost of energy (COE), grid connectivity status, location and type of software, which were implemented by various researchers for renewable energy generation based application, have been shown in Table 1.1 and Table 1.2. The detailed techno-economic feasibility of the stand-alone as well as grid connected hybrid power generation systems across the globe is presented in these studies. Renewable energy sources such as solar PV, wind turbine generator (WTG), biogas and biomass generator, and small-scale hydropower are commonly used at various locations to generate electricity.

1.2. Novelty and contributions

It can be inferred from the literature that various studies have been carried out in order to estimate the optimal configuration of various HRES based power systems, in terms of economics and reliability. The studies also indicate that the battery as an energy storage device has played a major role in the renewable generation based power system. However, from the analysis (Table 1), it can be seen that LA batteries have most commonly been used in these applications. Therefore, it is significant to perform economic investigations on the Li-ion battery for

Table 1.1
Critical review summary of various RESs configurations, battery types, cost of energy (COE), grid connectivity status, location and types of software used in power system.

Authors Name/Year	Location	System Type	Battery Type/Model	COE	On/Off Grid Status	Load type/ application	Software	Ref.
C. Phuralalpam, et al., 2018	Palari, Chhattisgarh & Hyderabad India	Photovoltaic (PV) array, Wind turbine generator (WTG)	Lead acid (L.A)battery	0.163 $/kWh	both	Small village load , residential building and IT business building load	HOMER	[9]
Y. Sawle, et al., 2018	Barwani, MP, India	PV/WTG/Bio/Diesel Generator (DG)	Flooded L.A battery	0.289 $/kWh	Off-grid	Residential / Village	HOMER and PSO	[10]
V. Tomar, et al., 2017	New Delhi, India	Grid connected PV	Flooded L.A (Hoppecke 12)	–	both	Slab wise and time of day (ToD) tariff structure	HOMER	[11]
M. Usman, et al., 2017	New Delhi, India	PV/WTG/DG	Flooded L.A Battery (Surrette 6CS25P)	8.84 Rs/kWh	both	Electrical load of faculty of Engineering and Tech., Jamia Millia Islamia, New Delhi	HOMER	[12]
H. Maammeur, et al., 2017	Chlef district, Algeria	PV	Without battery	0.6 $/kWh	On-grid	Family farms	HOMER	[13]
S. Rehman, et al., 2017	Bisha, Saudi Arabia	PV	With and without battery	–	On-grid	Grid	RET-Screen	[14]
B. K. Das, et al., 2017	Bangladesh	Biogas generator (BG)-PV-DG	L.A batteries (Trojan L16P)	0.28 $/kWh	Off-grid	Village	HOMER	[15]
M. K Shahzad, et al., 2017	Layyah in the Punjab, Pakistan	PV/Biomass	Li-ion batteries	5.51 PKR/kWh	Off-grid	Agriculture and residential load	HOMER	[16]
P. Peerapong, et al., 2017	Kohmak island, Thailand	PV/DG	Flooded L.A (Surrette 6CS25P)	0.374 $/kWh	Off-grid	Hotel, residential etc.	HOMER	[17]
R. Rajbongshi, et al., 2017	Jhawani village, Tezpur, India	PV/DG/BG	LA battery (Surrtte 6CS25P)	0.145–0.064 $/kWh	both	Jhawani village	HOMER	[18]
L.M. Halabi et al., 2017	Sabah, Malaysia	PV/DG	Fiamm type Battery	1.22-1.36 $/kWh	Off-grid	Remote areas of two island (Pulau Banggi and Tanjung Labian)	HOMER	[19]
A. H. Mamaghani, et al., 2016	Puerto Estrella, Colombia	PV/DG/WTG	L.A Batteries (4KS25P), Batteries (H3000)	0.414–0.868 $/kWh	Off-grid	Off-grid villages in Colombia	HOMER	[20]

Table 1.2
Critical review summary of various RESs configurations, battery types, COE, grid connectivity status, location and type of software used in power system.

Authors Name/Year	Location	System Type	Battery Type/Model	COE	On/Off Grid Status	Load type/application	Software	Ref.
A. Chauhan, et al., 2016	Chamoli, Uttarakhand, India	Mini Hydro Power-Biogas-Biomass-WTG-PV array	Lead Acid (L.A) battery	0.092 $/kWh	Off-grid	Village household load	HOMER	[21]
S. Singh, et al., 2016	Patiala, Punjab, India	PV-Biomass	Without battery	0.11 $/kWh	Both	Village load	ABC Algo. & HOMER	[22]
S. Singh, et al., 2016	Patiala, Punjab, India	PV-WTG-Biomass	L.A battery	0.017–0.181 $/kWh	Off-grid	Small village, situated in Patiala, Punjab India	ABC, PSO Algo. & HOMER	[23]
W. M. Amutha, et al., 2016	Kadayam, Tamil Nadu, India	PV/WTG/Hydro	Flooded L.A (Surrette$460)	0.111$ per- unit	Both	Household load in a village called Kadayam, Tirunelveli	HOMER	[24]
S. Salehin, et al., 2016	Kutubdia Island, Bangladesh	Solar PV-DG and WTG-DG	L.A battery (Surrte 6CS25P)	For PV-D 0.353 $/kWh For W-D 0.487 $/kWh	Off-grid	200 Households, 1000 People, Residents on island	HOMER & RETScreen	[25]
A. Ataei, et al., 2015	Tehran, Iran	Stand-alone hybrid PV-WTG system	SBV batteries	1.54 $/kWh	Off-grid	Office building located in Tehran, Iran	HOMER	[26]
M.A. Mohamed, et al., 2015	Saudi Arabia	PV/WTG/DG	Flooded type batteries	33.65 cents/kWh	Off-grid	Remote areas in Saudi Arabia	NPSP vs GA and HOMER	[27]
R. Sen, et al., 2013	Palari, Chhattisgarh, India	Hydropower, PV, WTG and Bio-DG	L.A battery (Surrette 6CS25P)	0.420 $/kWh	Off-grid	Off-grid remote village	HOMER	[28]
B. Dursun, et al., 2013	Edirne, Turkey	PV/WTG/Diesel System	Flooded L.A acid (Surrette 6CS25P)	0.364 to 0.679 $/kWh	Off-grid	50 houses at Edirne	HOMER	[29]
A. Hiendro, et al., 2013	Temajuk, Indonesia	PV/wind hybrid system	Flooded L.A battery	1.06 $/kWh	Off-grid	Temajuk, a small village in the West Kalimantan, Indonesia	HOMER	[30]
R. Kumar, et al., 2012	Jaipur, Rajasthan, India	Wind/PV/diesel hybrid energy system	L.A battery (Torjan L16P)	14.23 Rs/kWh	Off grid	Area of Jaipur	HOMER and BBO algorithm	[31]

renewable based generation applications considering their cost and important technical parameters. The main contribution of this work are outlined as follows:

- To design an optimal model of the standalone as well as grid connected microgrid system having PV-wind-diesel-biodiesel energy resources with battery bank in rural and urban scenarios.
- To assess the performance of the proposed microgrid system using realistic load profile, authentic resources data and real prices of components.
- To attain the optimal size of the components used in the proposed system with the least cost of energy and total net present cost of the system.
- To investigate the impact of the adoption of lead acid/lithium-ion battery storage on storage unit cost for different microgrid systems.
- To perform the techno-economic analysis and compare the performance of lithium-ion and lead-acid batteries in the proposed system.

The article is arranged as follows: the modelling of microgrid components and their economic parameters are given in Section 2. Section 3 deals with the resources data and load profile of the selected locations. Simulation results, related discussions and conclusion have been represented in Sections 4 and 5, respectively.

2. Microgrid components modelling and economic parameters

A microgrid is a cluster of distributed energy resources (DERs) such as micro-turbines, diesel/biogas generators, fuel cells, wind generators, photovoltaic systems, with energy storage and interconnected loads, located close to the consumer, that acts as a single well-regulated entity with respect to the grid [32]. The schematic of a microgrid system having different components is presented in Fig. 1.

2.1. Solar photovoltaic generation

A solar-photovoltaic (PV) system converts solar energy into electrical energy with the help of solar cells. The power output depends on atmospheric conditions and geographical location [22]. The PV array power output (P_{SPV}), can be calculated as:

$$P_{SPV} = G_{PV} f_{PV} \left(\frac{I_T}{I_{T,STC}}\right)[1 + \alpha_P (T_C - T_{C,STC})] \quad (1)$$

where I_T is the solar radiation incident on the PV array in kW/m^2, G_{PV} denotes PV rated capacity in kW, f_{PV} is the PV derating factor, $I_{T,STC}$ represents the solar incident radiation at STC (standard test conditions) (1 kW/m^2), T_C is the cell temperature of the PV in °C, α_P is the temperature coefficient of power (%/°C), $T_{C,STC}$ is the cell temperature under STC (25 °C) of the PV [9,15,22,23]. Further, the output-generated power of the PV is also reliant on the wind speed and cell temperature around the panel. The energy balance for PV module can be calculated as follows:

$$(\tau\alpha) I_T = I_T \eta_{PV} + U_L (T_C - T_a) \quad (2)$$

where $(\tau\alpha)$ denotes the effective absorptance-transmittance of the PV panels, η_{PV} is the efficiency of PV array, U_L denotes the heat transfer coefficient from the surroundings, T_C and T_a denote the cell temperature and ambient temperature, respectively [15]. The above equation can be rewritten as follows:

$$T_C = T_a + I_T \left(\frac{\tau\alpha}{U_L}\right)\left(1 - \frac{\eta_{PV}}{\tau\alpha}\right) \quad (3)$$

As per the manufacturers specification, the value of $(\tau\alpha)$ used in (3) can be measured at Nominal Operating Cell Temperature (NOCT) with other conditions such as at 20 °C ambient temperature, 0.8/kW/m^2 of solar irradiation, and at no load conditions (i.e.η_{PV} = 0) [22,23]. By

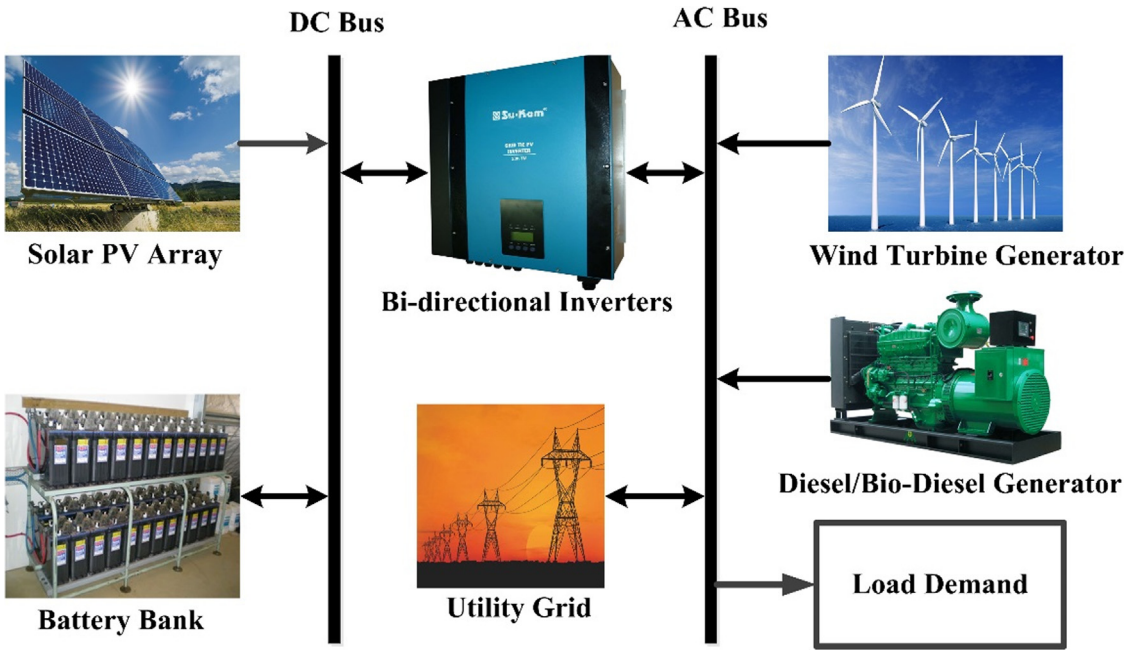


Fig. 1. Block diagram of a microgrid having a hybrid supply of PV/Battery/Wind/Diesel.

using these values, (3) becomes:

$$\frac{\tau\alpha}{U_L} = \frac{T_C{}^{NOCT} - T_a{}^{NOCT}}{I_T{}^{NOCT}} \tag{4}$$

where $I_T{}^{NOCT}$ indicates irradiation of the solar at NOCT (0.8/kW/m^2), $T_{C,NOCT}$ denotes the NOCT, $T_a{}^{NOCT}$ denotes the atmospheric temperature at NOCT.

HOMER takes the value of ($\tau\alpha$) as 0.9. The cell temperature of the PV panel is finally calculated by using ($\tau\alpha$) and can be expressed as follows [15]:

$$T_C = T_a + I_T\left(\frac{T_{C,NOCT} - T_{a,NOCT}}{I_{T,NOCT}}\right)\left(1 - \frac{\eta_{PV}}{\tau\alpha}\right) \tag{5}$$

A flat plate, multi-crystalline solar module type PV panel is taken in this study. The derating factor, efficiency at STC (%) and the lifetime of the PV panels are taken as 80%, 14.5% and 20 years, respectively [9]. The factors such as snow cover, shading, dust particles, aging, high temperature, wiring losses, which cause variation in the output power of the PV array from its rated capacity are combined to give a derating factor [9]. The capital cost, replacement cost and operation & maintenance (O&M) cost of the PV are taken as 840 $/kW, 800 $/kW and 10 $/kW/year, respectively [33]. The capital cost includes the charges for delivery, installation, wiring and mounting hardware.

2.2. Wind turbine generation model

A WTG produces electricity by utilizing mechanical energy present in the wind. The WTG power output (P_{wt}) can be calculated as follows:

$$P_{wt}(v) = \begin{cases} \frac{P_r(V - V_{cin})}{V_{rat} - V_{cin}} & V_{cin} \leqslant V \leqslant V_{rat} \\ P_r & V_{rat} \leqslant V \leqslant V_{CO} \\ 0 & V \leqslant V_{cin} \text{ and } V \geqslant V_{CO} \end{cases} \tag{6}$$

where P_r denotes the rating of a single WTG, V_{cin} and V_{CO} are the cut-in speed and cut-out speed, respectively, 'V' represents the wind speed at desired height, V_{rat} is the rated wind speed [15].

The speed of the wind at the particular hub height mainly depends upon the geographic location, which can be denoted as represented in (7),

$$V = V_{ref}\left(\frac{H}{H_{ref}}\right)^{\alpha} \tag{7}$$

where 'V' represents the wind velocity at a height of 'H', V_{ref} designates the wind velocity calculated at the reference height (H_{ref}). Also, the exponent 'α' depends on wind speed, surface roughness, time of day, temperature and on season. The value of α is normally considered as 0.142 for consistent wind flow, but its value can vary from 0.4 for rough surface conditions in temperate latitudes, to 0.05 for smooth surface conditions in the tropics [15]. The variation in the air density can also influence the actual power output of the WTG. HOMER multiplies the air density ratio and power value, which is determined from the power curve in order to standardise it to realistic conditions, as denoted by equation shown below:

$$P_{wt} = \left(\frac{\rho}{\rho_O}\right)P_{wt,(STP)} \tag{8}$$

where $P_{wt,(STP)}$ denotes the WTG power output at standard temperature and pressure (STP), ρ is the actual air density, ρ_O is the air density at STP (1.225 kg/m^3) [15,23].

Two generic WTGs of 1 kW and 10 kW rating capacities are considered for this study. The capital cost, replacement cost and O&M cost of the 1 kW WTG are considered as 2300 $/kW, 1500 $/kW and 2 $/kW/year, respectively [23]. The lifetime, overall efficiency and the hub height of the WTG are taken as 20 years, 26% and 50 m, respectively [23]. The cut-in speed (V_{cin}), cut-out speed (V_{CO}) and rated wind speed (V_{rat}) of 1 kW are taken as 3 m/s, 20 m/s and 11 m/s, respectively [23]. Similarly, the capital cost, replacement cost and O&M of the 10 kW WTG are taken as $11,539, $11,539 and 230 $/year, respectively [9]. The lifetime and the hub height of the WTG are considered as 20 years and 25 m, respectively [9]. The cut-in speed (V_{cin}), cut-out speed (V_{CO}) and rated wind speed (V_{rat}) of 10 kW WTG are taken as 3 m/s, 24 m/s and 12 m/s, respectively [9]. The capital cost of the WTG comprises tower cost, turbine and rotor cost, control mechanism, wiring and installation charges.

2.3. Diesel generator model

A diesel generator (DG) is operated in the microgrid system when there is insufficient energy generation from RER based generators or

Table 2
Battery cost and technical parameters.

Parameters	Battery Type/Model	
	Lead-acid Battery (Amaron) Flooded LA	Li-ion Battery Smart Battery (LiFePO4 type)
Nominal cell voltage (V) and maximum capacity (Ah)	12 V, 83.4 Ah	6 V, 167 Ah
Nominal capacity	1 kWh	1 kWh
String size	1	2
Cycle life @ maximum DOD	800	3200
Float life	4 years	12 Years
Replacement timeframe (years)	2.5–4.5	10.5–12.5
Initial state of charge (SOC) in (%)	100	100
Minimum SOC in (%)	40	10
Round-trip efficiency (%)	80	90
Cost ($/kWh)	124	630
Replacement cost ($/kWh)	124	600
Operation and maintenance ($/kWh/year)	10	0
Delivery, installation and other (Labour and structure) charges	$ 30 per kW	$ 20 per kW
References	[9,36,37]	[8,36]

during peak load demands to provide continuous power supply. The fuel consumption rate for electricity generation with a diesel/biodiesel generator can be given as:

$$F = F_{0,DG} Y_{DG} + F_{1,DG} P_{DG} \tag{9}$$

where F_0 and F_1 are the intercept coefficient and slope of the fuel curve, Y_{DG} and P_{DG} are the rated capacity and maximum electrical output of the generator, respectively [9,15].

Currently, the use of biodiesel an environment friendly fuel, has received a lot of attention because biodiesel can be used in a diesel engine without major engine modifications [34]. Many researchers have concluded that biodiesel is a promising alternative fuel for diesel engines [34,35]. Therefore, a generic 1 kW genset, having capital and replacement cost of 385 $/kW, has been taken from the HOMER resources library for the simulation for both diesel and biodiesel generators. The O&M cost of the generator per kW is considered as 0.15 $/h. The lifetime of the generator depends upon its operating hours and is considered as 15,000 h [9]. In this study, 1 $/litre and 0.53 $/litre prices have been considered respectively for the fuels of diesel and biodiesel generators (BDG) [9].

2.4. Power converter

A bi-directional converter has been employed to link the AC and DC buses, which are connected to the other components (PV modules, WTG, DG, battery and load) of the microgrid system as displayed in Fig. 1. The size of the converter mainly depends on peak load demand (P_L^{max}) [15]. The rating of the inverter (P_{inv}) can be obtained by the expression given below:

$$P_{inv} = P_L^{max} / \eta_{inv} \tag{10}$$

where η_{inv} indicates inverter efficiency [22,23].

Here, both capital and replacement costs of the converter have been considered to be 127 $/kW. The O&M cost of the converter is taken as 1 $/kW/yr. The lifespan of the converter is taken as 15 years (in cases 2, 3 & 4) and 20 years (in case 1) with round trip efficiency of 90% in this study [22,23].

2.5. Battery modelling

Batteries in a microgrid system are used as the storage medium. The battery bank stores the surplus energy during low power demand and discharges its stored energy during peak demand or while insufficient electricity is supplied from renewable generators. The battery energy can be calculated as given below:

$$B_{bat} = B_{bat.0} + \int_0^t V_{bat} I_{bat} dt \tag{11}$$

where $B_{bat.0}$ denotes the initial battery charge, V_{bat} and I_{bat} are the voltage and current of the battery respectively [15]. The accurate estimation of the state of charge (SOC) is required to calculate the appropriate value of the battery energy. The SOC of the battery varies with respect to time, which can be expressed by (12):

$$\frac{SOC(t)}{SOC(t-1)} = \int_{T-1}^{T} \frac{P_{bat}(t)\eta_{bat}}{V_{bus}} dt \tag{12}$$

where V_{bus} is the bus voltage of the microgrid system, η_{bat} denotes the battery round-trip efficiency. The power supplied by the battery is symbolized by '$P_{bat}(t)$' [23]. Hence, the battery round-trip efficiency is expressed as follows:

$$\eta_{bat} = \sqrt{\eta_{bat}^{c} \eta_{bat}^{d}} \tag{13}$$

where η_{bat}^{d} and η_{bat}^{c} denote the discharging and charging efficiency of the battery, respectively [23]. The maximum value of SOC is denoted by 'SOC_{max}' which is equal to the aggregate capacity of the battery bank (C_n(Ah)). It can be expressed as:

$$C_n(Ah) = \frac{N_{bat}}{N_{bat}^{s}} C_b(Ah) \tag{14}$$

where N_{bat} denotes the total number of batteries, C_b(Ah) represents the single battery capacity, N_{bat}^{s} is the number of batteries connected in series. The minimum limit of the discharge of a battery bank is known as 'SOC_{min}'. This limit mainly depends on the type of battery and its usage in the battery bank [15]. The batteries are connected in series to obtain the required bus voltage. The number of batteries connected in series to attain the required voltage is given as:

$$N_{bat}^{s} = \frac{V_{bus}}{V_{bat}} \tag{15}$$

where V_{bat} denotes the voltage rating of a single battery. Further, the value of the maximum charge/discharge power of a battery at any time is stated as follows:

$$P_{bat}^{max} = \frac{N_{bat} V_{bat} I_{bat}^{max}}{1000} \tag{16}$$

where I_{bat}^{max} represents the maximum charging current of the battery in amperes [23]. The technical details and cost of the batteries used in this study are given in Table 2.

2.6. Grid

One of the microgrid systems under study is partially connected to the grid. Here, the power is available for a few hours in a day due to unreliable grid supply. The load shedding pattern of the village, which is connected to an unreliable grid is displayed in Fig. 2. An average availability of electricity in a day is 12 h in this village, which is represented by black colour in the figure, while the green colour presents the unavailability of the electricity. In Chhattisgarh, India, the per-unit cost of the electricity varies between $0.05 to $0.10 for domestic usages [9]. The electricity price of the grid is taken as 0.08 $/kWh in the simulation for this case study.

2.7. Economic modelling

The economic feasibility of the proposed microgrid systems under study has been evaluated on the basis of the per-unit cost of energy (COE), and the total net present cost (TNPC) of the whole system. A brief introduction about these parameters is given below:

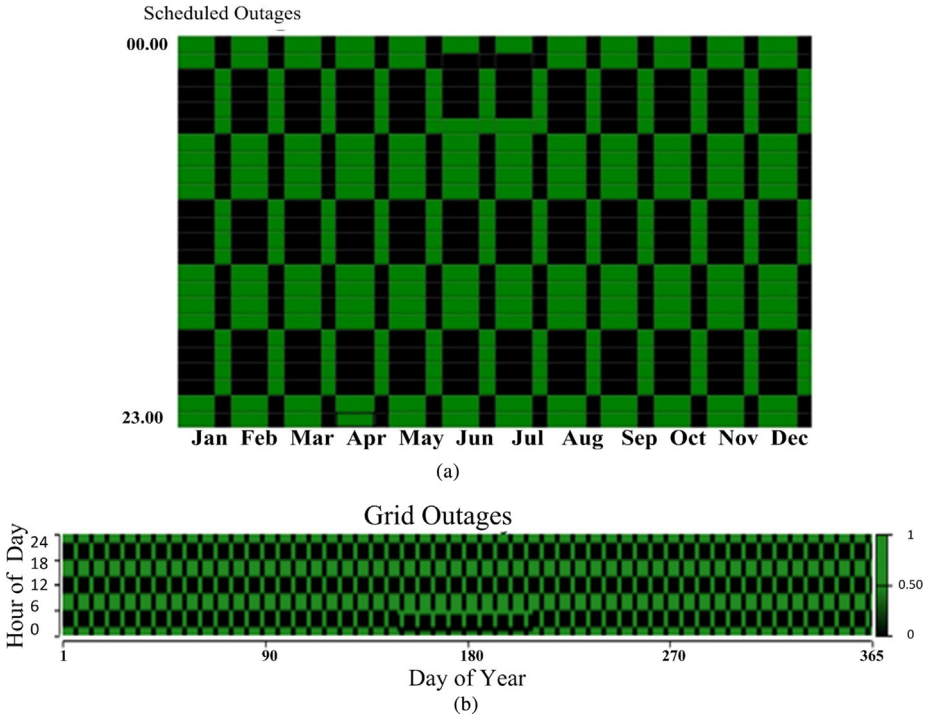


Fig. 2. Load shedding pattern in the village interconnected with an unreliable grid (a) Hours (on y-axis) vs months of the year (on x-axis). (b) Hours of the day (on y-axis) vs days of the year (on x-axis).

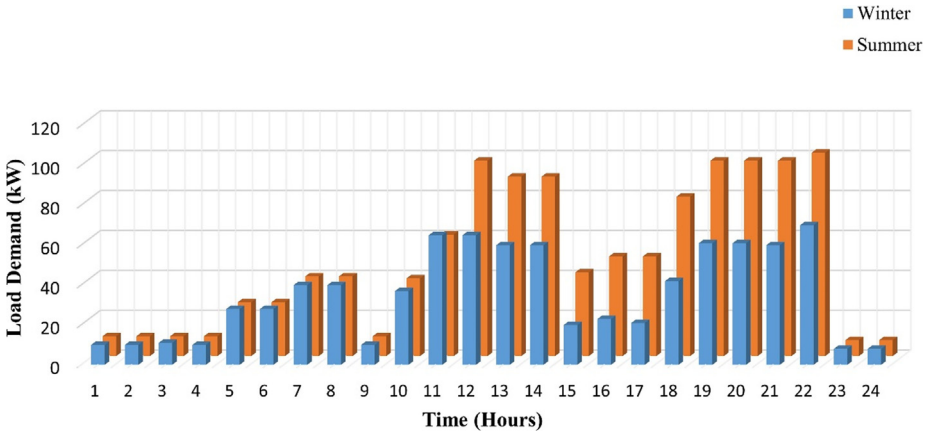


Fig. 3. Daily load profile (winter and summer (kW)) of a small village, situated in Patiala, Punjab, India.

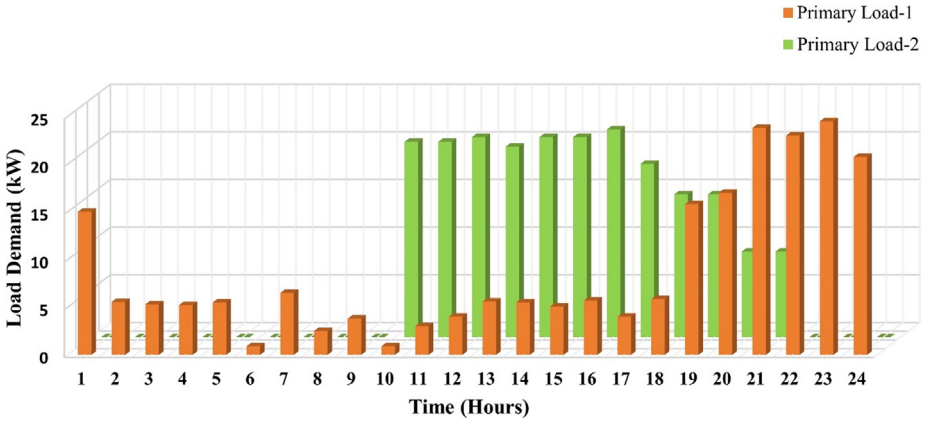


Fig. 4. Everyday load profile of a specific village, Palari in Chhattisgarh, India.

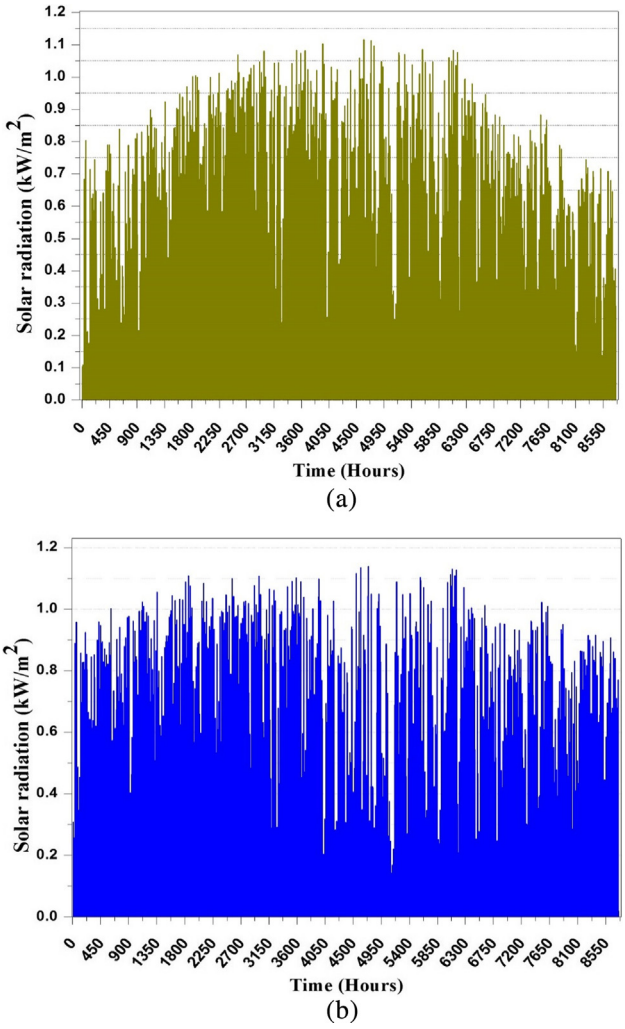


Fig. 5. Annual data of solar GHI (Global Horizontal Irradiance) (a) a small village, situated in Patiala, Punjab, India, (b) a village at Palari in Chhattisgarh.

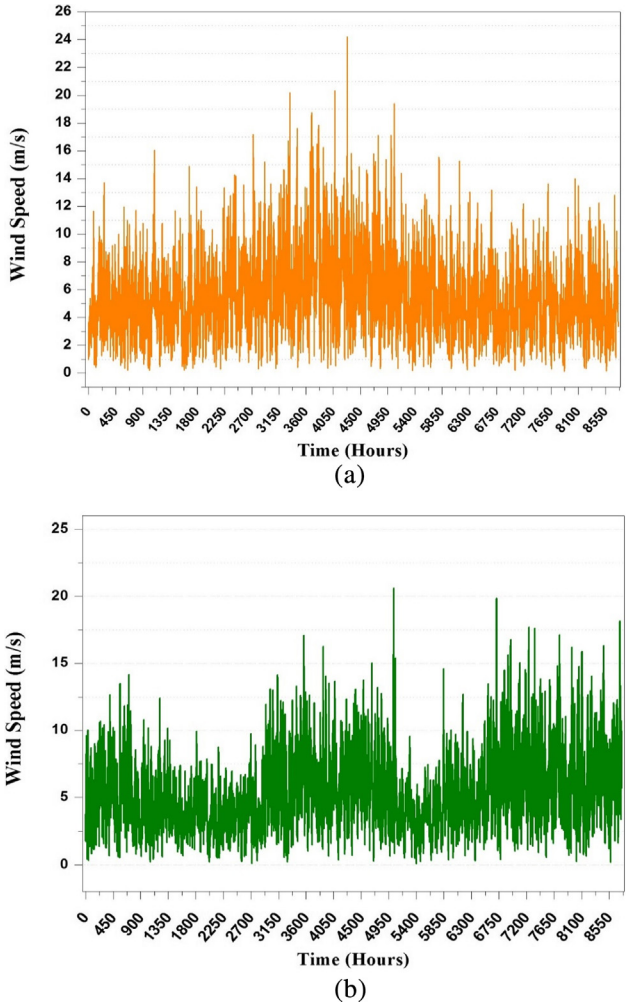


Fig. 6. Average wind speed data for one year (a) a small village, situated in Patiala, Punjab India, (b) India's largest wind farm (Muppandal Wind Farm).

2.7.1. Cost of energy

For the economic assessment, the per-unit COE is used for an integrated hybrid energy system. The COE is defined as the ratio of system components annual cost by the total energy generated [38]. This energy generated is the actual demand (excluding excess energy) supplied and is represented by (17):

$$COE = \frac{C_A}{E_S} \tag{17}$$

where 'E_S' is the yearly energy supplied and 'C_A' is the total annual cost [15]. The annual cost is comprised of annual capital cost (C_{A_CC}), annual operation and maintenance cost ($C_{A_O\&M}$), and annual replacement cost (C_{A_rep}). This cost can be calculated by (18) as follows:

$$C_A = C_{A_CC} + C_{A_rep} + C_{A_O\&M} \tag{18}$$

2.7.2. Net present cost

The TNPC of a system can be defined as the current value of all the initial capital costs, costs of the replacement, cost incurred in O & M, and fuel costs minus the current value of all the revenues. The revenues include salvage values of system components for the entire lifespan of the project. The TNPC is expressed as:

$$NPC = \frac{C_A}{CRF(i,n)} \tag{19}$$

where CRF(i,n) represents the capital recovery factor, 'n' and 'i' denote the number of years and the real annual interest rate, respectively [15]. Further,

$$CRF(i, n) = \frac{i(1+ i)^n}{(1+ i)^n-1} \tag{20}$$

where, $i = \frac{i_{nom} - f}{1+f}$

where, 'i_{nom}' represents the nominal interest rate and 'f' denotes the yearly inflation rate. In the present study, the project lifespan has been taken 20 years with an annual inflation rate of 2% and discount rate of 8% [23].

3. Resources data and estimated load profile at the selected location

Four different load profiles are considered in this study. The first and second load profiles belong to two different villages representing a rural scenario. The third one constitutes an urban building, and the last profile considered is the load of a commercial establishment in an urban area. All load profiles have been taken from recently published literature related to realistic rural and urban Indian scenarios by following expert opinions and personal judgments [9,23,28].The detailed discussion about the various load profiles used in microgrid simulation are given in the following sub-sections:

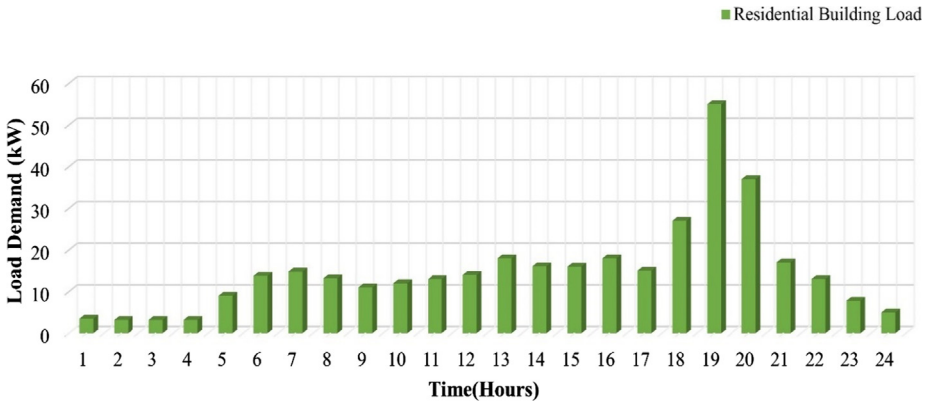


Fig. 7. Daily load profile of a residential urban building or a housing complex in India.

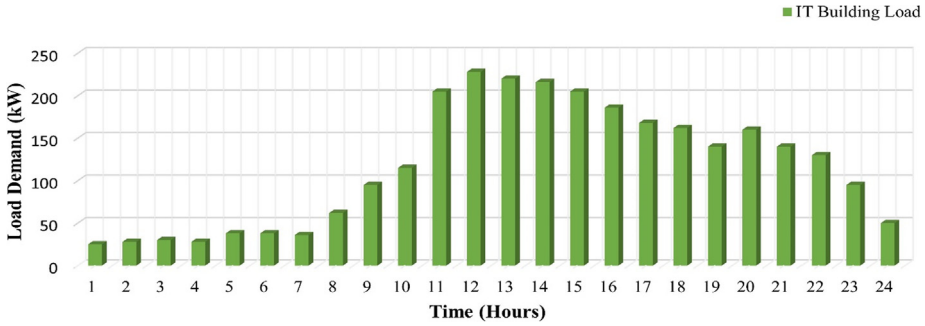


Fig. 8. Average daily load profile of an IT business organisation at Hyderabad, India [9].

3.1. Case-1 Village load (rural scenario)

In this scenario, a small village is selected which is located at latitude of 30°26′N and longitude of 76°12′E in Patiala, in Punjab state of India. The village household loads have a peak load demand of 102 kW with a load factor of 0.406 as shown in Fig. 3 [23]. Two types of load profiles have been assessed individually for two different seasons prevalent in this area, i.e., winter (October to April) and summer (May to September) considering typical appliance use by the families, supply capability and commercial activities. An in-depth description about the load estimation considering the types of load, appliances used and their wattage for the studied area is given in Ref. [23].

The locality selected for this microgrid study has good solar and wind resources throughout the year. To produce actual modelling results of the planned microgrid system, the realistic data of solar and wind resources, which are available at NASA (National Aeronautics and Space Administration) surface meteorology and solar energy database are used in the simulation. The annual average solar radiation has been found to be 5.44 kWh/m^2/day and annual average wind speed is 5.9 m/s for the selected location as displayed in Fig. 5(a) and Fig. 6(a), respectively [23,39].

3.2. Case-2 Load of the village partially connected to an unreliable grid (rural scenario)

A village situated at latitude of 19°365′N and longitude of 81°672′E at Palari in Chhattisgarh, India is selected for this scenario. The village load has been divided into three essential clusters by considering appliance usage patterns of families, supply limits and commercial activities as given below [9,28]:

1. **Primary Load-1**: The first type of load comprises the domestic load, medical centre and school load demand. The total demand of the load is approximately 222 kWh/day and 51.2 kW peak demand with a load factor of 0.181 as shown in Fig. 4.
2. **Primary Load-2**: This load includes the demand of shops, local business, a community centre and small manufacturing units. It is approximately 212 kWh/day and 39.4 kW peak with a load factor of 0.224 as shown in Fig. 4.
3. **Deferred Load**: The agricultural load of the village makes up this type of load. The scaled annual average deferred load is 58.6 kWh/day and has a peak load of 68.6 kW. It has a storage capacity designed for 30 kW, which is connected on the AC side.

An in-depth description of the load estimation considering the type of load, appliances used and their wattage for the studied area is given in Refs. [9,28]. For all other cases, the realistic data of solar and wind resources, which are available at NASA surface meteorology and solar energy database are used in the simulation. The wind resource data of India's largest wind farm (Muppandal Wind Farm), which is situated near Kudankulam in Tamil Nadu (India) have been scaled for the cases 2,3 and 4 in this study. The yearly average solar radiation is found to be 5.17 kWh/m^2/day at the selected location and yearly average wind speed is 5.45 m/s as displayed in Fig. 5(b) and Fig. 6(b) [9,28,39].

3.3. Case-3 Community/building load profile (urban scenario)

A multi-storey housing complex or a residential urban building is the next type of load considered. As measured electricity, consumption data for this type of load in India are relatively difficult to find, the hourly load profile is generated by giving average daily power demand data as an input to HOMER. A comprehensive study is conducted for computing this value. The building is considered to be ten storied, with each level having four apartments of 2–3 rooms each. The number of household appliances and the energy consumption are usually higher in urban areas as compared to rural. The electricity requirement for the building household loads has an average total demand of 358.80 kWh/day with a load factor of 0.20, are considered in this study. The average

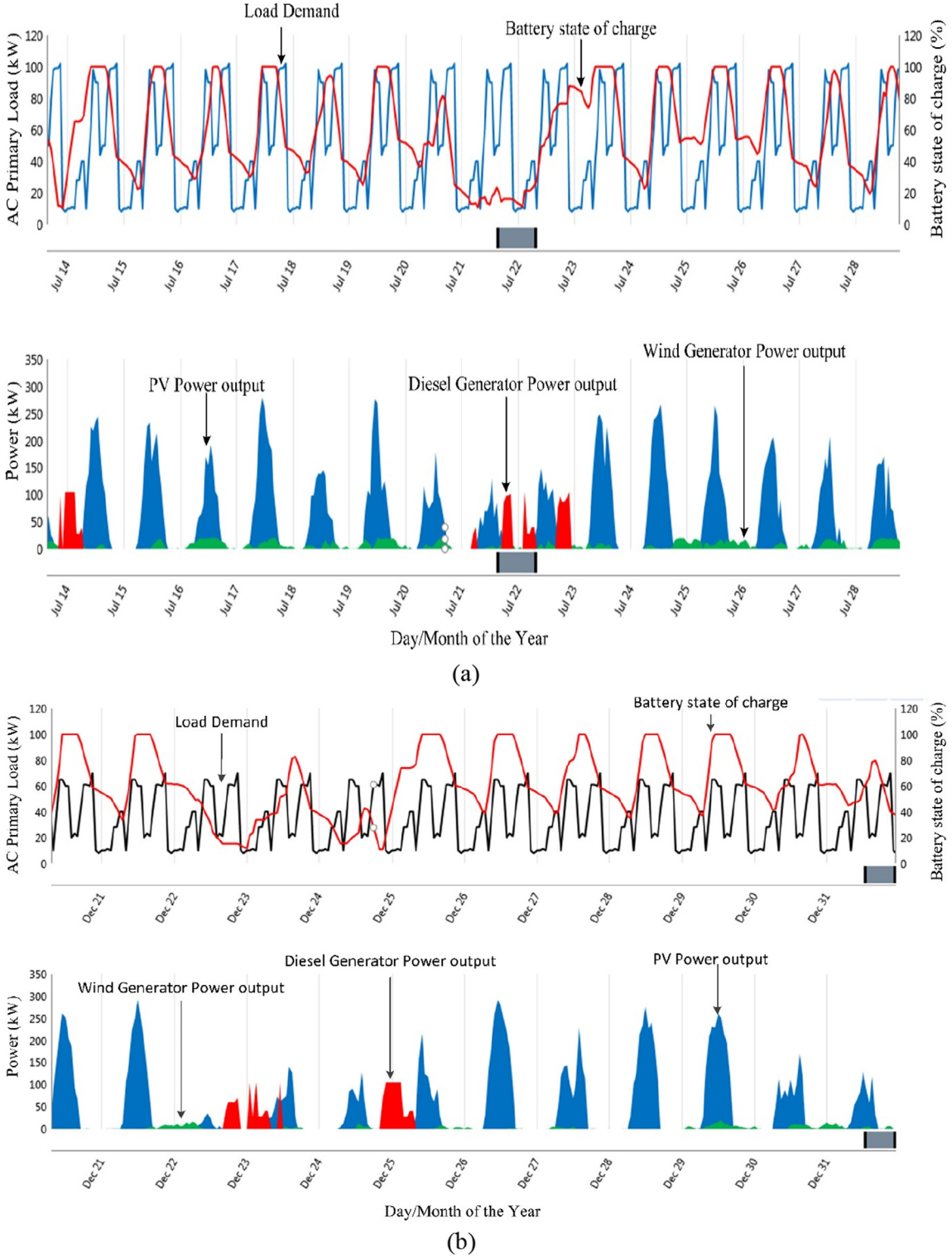


Fig. 9. Energy generation from different sources and battery SOC for the load demand of (a) two weeks in July (b) 12 days in December.

electricity consumed in number of hours per day by each appliance throughout the year and average daily demand of the building is fed into HOMER, which generates an hourly load profile as shown in Fig. 7. A detailed description about the various types of electrical loads in each household or apartment and appliances are given in Ref. [9].

3.4. Case-4 Commercial establishment/building (urban scenario)

In this case, the load profile of a commercial establishment (IT building) situated in Hyderabad, India is considered. The complete detail of the measured data for load profile is given in Ref. [9]. The business organisation incorporates a total area of 3.81 sq km. The electrical loads in the facility are generally categorized into priority and deferrable loads. The average daily load pattern of the priority load is

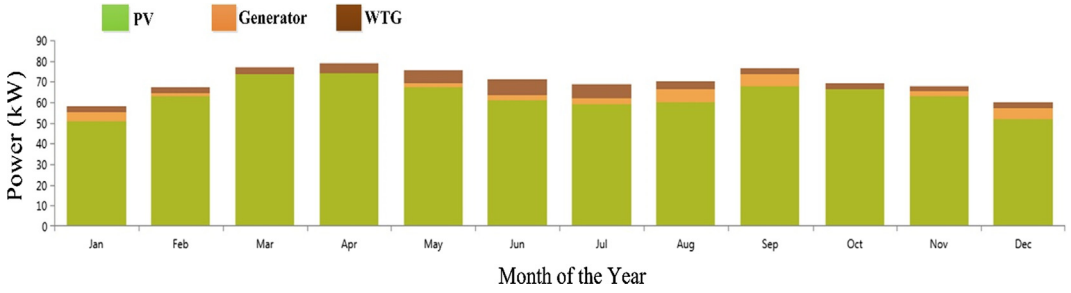


Fig. 10. Monthly average electricity generation from different sources.

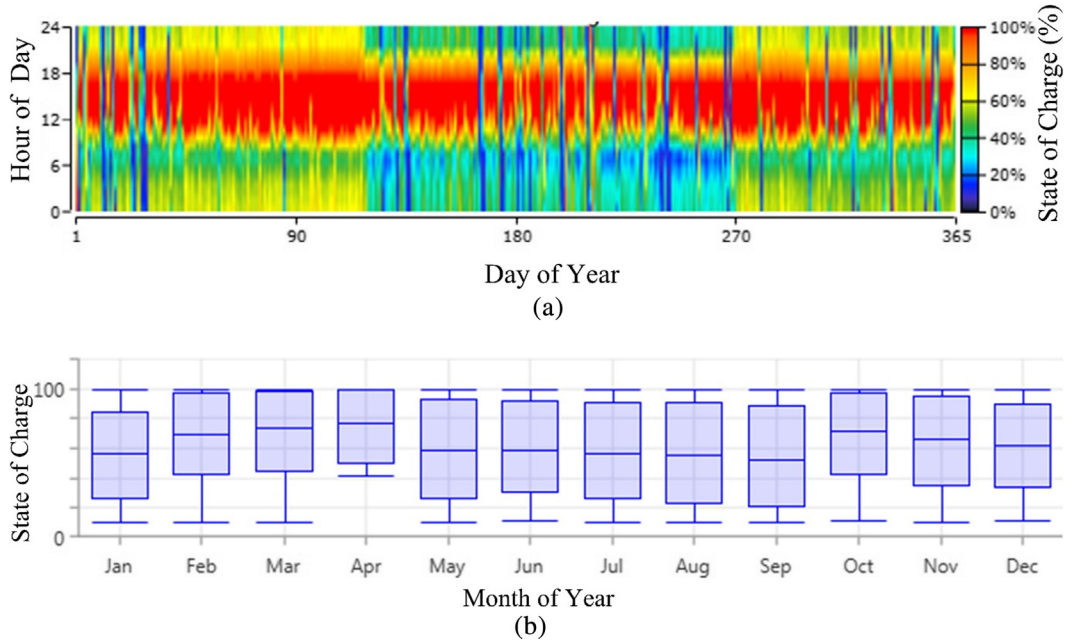


Fig. 11. Lithium-ion battery SOC (a) Hours of the day vs. day the of year, (b) SOC vs. month of the year.

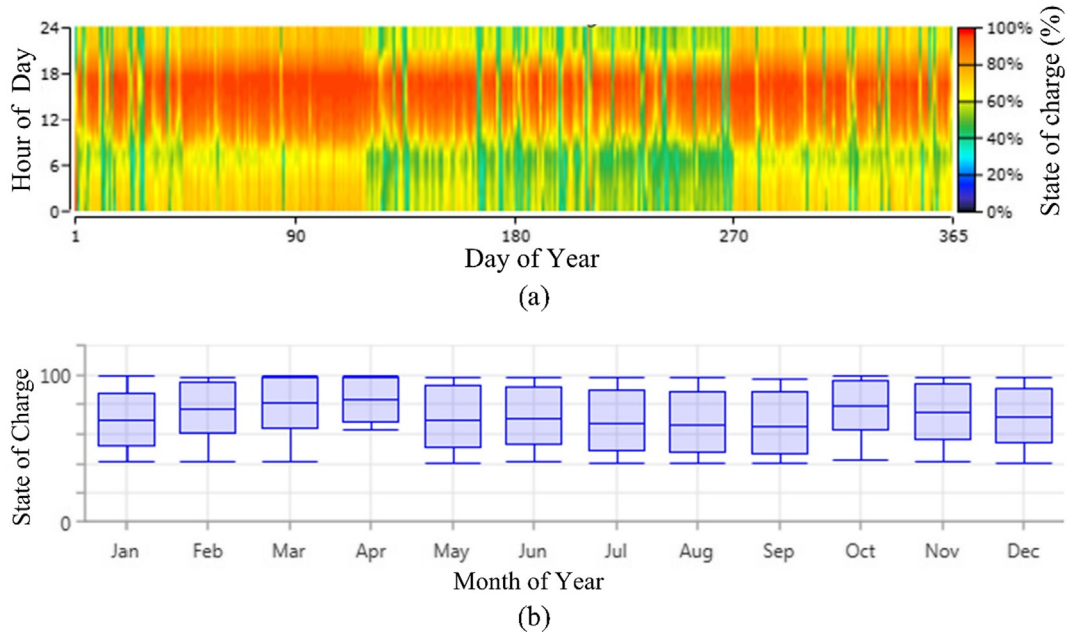


Fig. 12. Lead-acid battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

Table 3
Cost summary for different components used in case-1.

Components		Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV		252,000	0.00	41055.61	0.00	0.00	293055.61
Wind Generator		46,000	0.00	547.41	0.00	0.00	46547.41
Battery	Li-ion	510,900	267861.98	0.00	0.00	−139831.55	638930.42
	LA	165,664	457757.89	182864.30	0.00	0.00	806256.01
Diesel Generator	Li-ion	40,425	0.00	736.95	98601.19	−9764.51	129998.60
	LA	40,425	0.00	749.26	96731.41	−9614	128290.98
Converter	Li-ion	13134.48	0.00	1415.34	0.00	0.00	14549.82
	LA	13003.62	0.00	1401.24	0.00	0.00	14404.86
Total System	Li-ion	862459.48	267861.98	43755.30	98601.19	−149596.09	1123081.8
	LA	517092.62	457757.89	226587.81	96731.41	−9614.70	1288555.0

displayed in Fig. 8. The seasonal variation in the priority load throughout the months of the year is also shown in the same figure. The building has an average daily demand of 2.8 MWh with a peak demand of 245 kW [9].

4. Simulation results and discussion

This section describes the performance of the batteries in various microgrid systems having different load scenarios. The proposed microgrid system comprises different power generators (PV, WTG, and DG/BDG), converters and batteries for energy storage. The systems have been developed and investigated using HOMER-2018 (13.11.3) Pro edition software. The studies have been carried out by taking realistic electricity load profiles of the rural and urban load scenarios. The cost parameters are taken in US dollar ($) in all the simulations and 1 USD is assumed equal to 64 Indian rupees (INR) [40]. In all cases, the lifespan of PV panels and WTG are taken equal as the lifetime of the project, therefore, no replacement is needed. Some components which can be used even beyond the lifespan of the project have negative values shown against them in the table of the cost representing their salvage cost. Battery life is dependent on usage taking into account whether LA or Li-ion type is used. The different case studies conducted under realistic load scenarios to perform the comparative analysis between Li-ion and LA batteries are as given below:

4.1. Case-1: PV/WTG/DG/Battery

The optimal results of an AC microgrid system having PV, WTG and DG are shown in Figs. 9–12. The viable and optimum solutions are ranked on the basis of TNPC, COE and the best result in case-1 comprises a system with 300 kW of PV, twenty units of 1 kW WTG each, 105 kW DG and a 103 kW converter unit. The microgrid system having Li-ion battery as a storage medium requires 786 units of 1kWh batteries, whereas the system having LA battery requires 1336 units. The combined dispatch (CD) strategy has been used in this simulation. The calculated energy cost for a continuous and reliable power supply from the microgrid system comes out to be 0.22–0.25 $/kWh. Most of the power demand of the load in the proposed microgrid systems is fulfilled with the help of RER based generators (PV and WTG), which results in reduction of CO_2 emissions by 18,496 kg per annum in this case study.

The response of the designed microgrid system in the form of power generated by the various generators, load power demand and battery SOC in the months of July (summer) and December (winter) are shown in Fig. 9(a) and (b), respectively. To validate the optimal operation of the proposed system for one year, a total period of two weeks in a month has been selected. In these figures, the upper section represents the output waveforms of total primary load and SOC of the battery, whereas the lower section represents the power generated by PV, WTG and DG. The obtained results clearly show that the generation from the RESs are capable of assisting the loads and it can keep the battery SOC within the limits. The DG is only operated in the absence of wind and solar energy for reliable operation. Hence, the proposed microgrid system provides reliable power supply and fulfils total energy demand with the help of wind/solar/diesel generators and batteries.

Fig. 10 shows the monthly average energy balance for one year from different generators of the microgrid system. The annual average electricity production by the PV is higher than WTG due to its higher rating in the microgrid system. The power generated by the renewable generators (solar and wind) are dependent on the wind and solar resources. The results show that a high level of solar and wind power is produced due to good natural resources at the selected location. When the power generated by PV panels, and WTG is not sufficient, then the remaining demand is fulfilled by the DG.

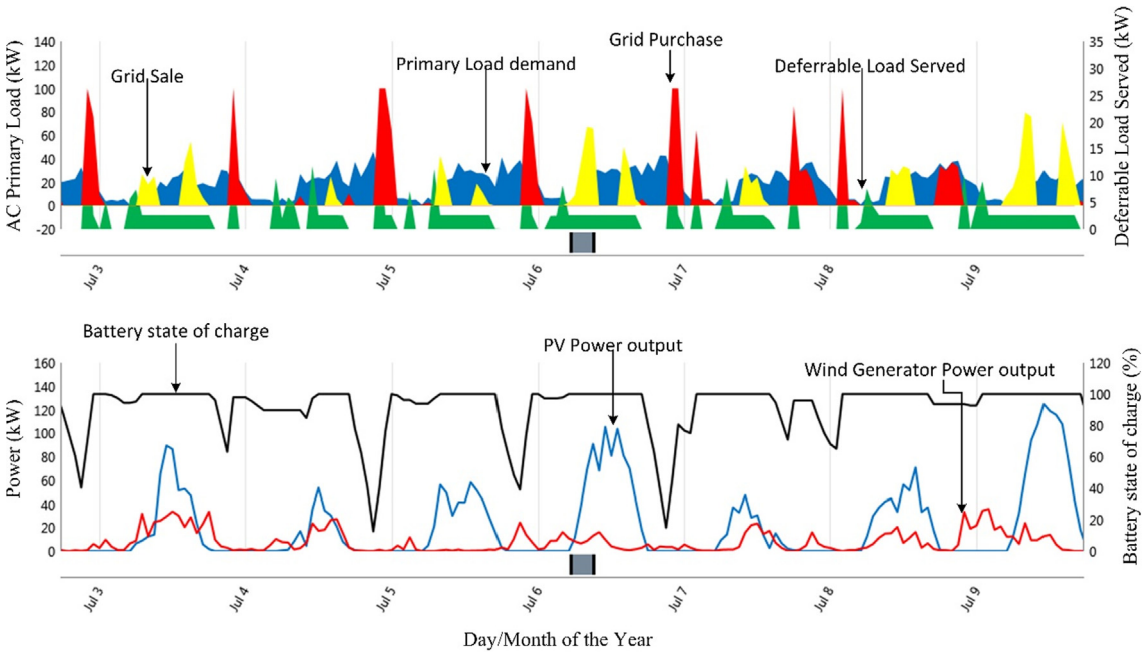
The battery unit aids the intermittency of RER generators by storing the energy generated and making it available upon demand. Figs. 11 and 12 give the average SOC of the Li-ion and LA battery unit for each month of the year. The results show that the SOC is kept within the specified limit for Li-ion (10–100%) and LA (40–100%) batteries, respectively. Also, the deployment of the battery bank increases during summer. The maximum utilization of the battery storage is observed in the month of January, and from May to September.

Table 3 demonstrates the complete and detailed analysis of the various cost components of the optimal microgrid system. The COE is the average cost for each unit (kWh) of generated electricity by the system, which mainly depends upon the net present cost of the system. The net present cost of the whole setup having Li-ion batteries is around $1.12 million and for the system having LA batteries is around $1.29 million. Similarly, the microgrid system having Li-ion batteries provides reliable power supply at 0.225 $/kWh compared with LA batteries at 0.258 $/kWh. The lifetime of DGs hinge on the operating hours instead of life in years. In this case, the operating hours of the DG are low, so no replacement is required. The main component, which requires replacement, is the battery bank. It can be observed that the battery unit has the maximum capital cost followed by the PV system. The O&M cost is led by the LA battery bank among other units. The set of batteries needs replacement according to their lifetime and usage. Li-ion batteries consist of 56% of the total cost of the system, whereas LA batteries consist of 62% of the total cost of the system. For the other components, solar PV contributes 26%, WTG 4% and DG 11% of the total lifetime costs.

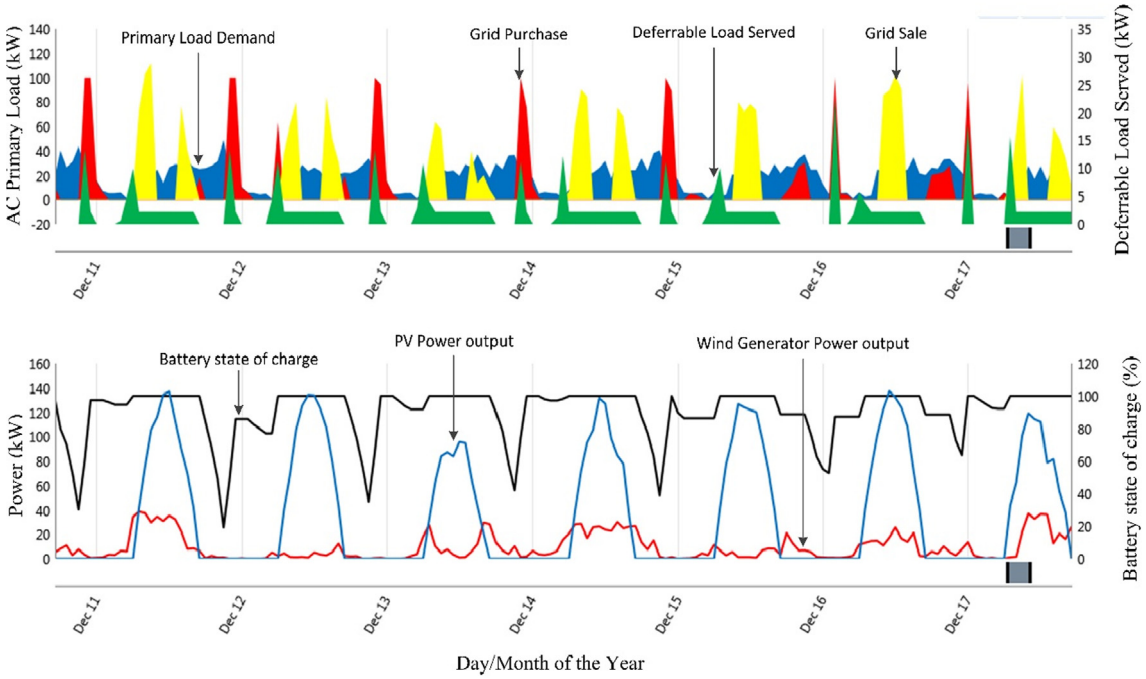
4.2. Case-2: PV/WTG/Battery connected with unreliable grid

The result of optimization for an AC microgrid system having partial grid connectivity for the village load is shown in Figs. 13–16. As described in Section 2.6, the village is connected to an unreliable grid that can only receive electric power supply for an average of 12 h in a day. On basis of grid power obtainability, HOMER gives the optimized results of the village AC microgrid.

The best optimised result in case-2 is ranked on the basis of TNPC and COE, and which recommends 1 unit of 150 kW of PV, four units of 10 kW WTG (G10) each, and 100 kW converter unit for its system. The



(a)



(b)

Fig. 13. Energy generation from different sources and battery SOC for the load demand of (a) the first week of July, (b) the second week of December.

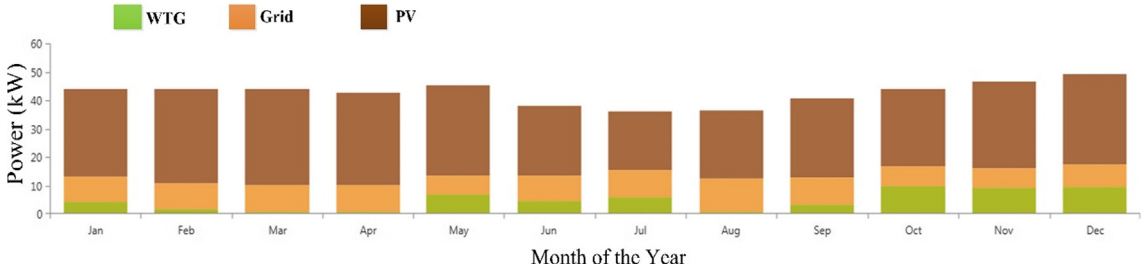


Fig. 14. The monthly average electricity generation from different sources in a year.

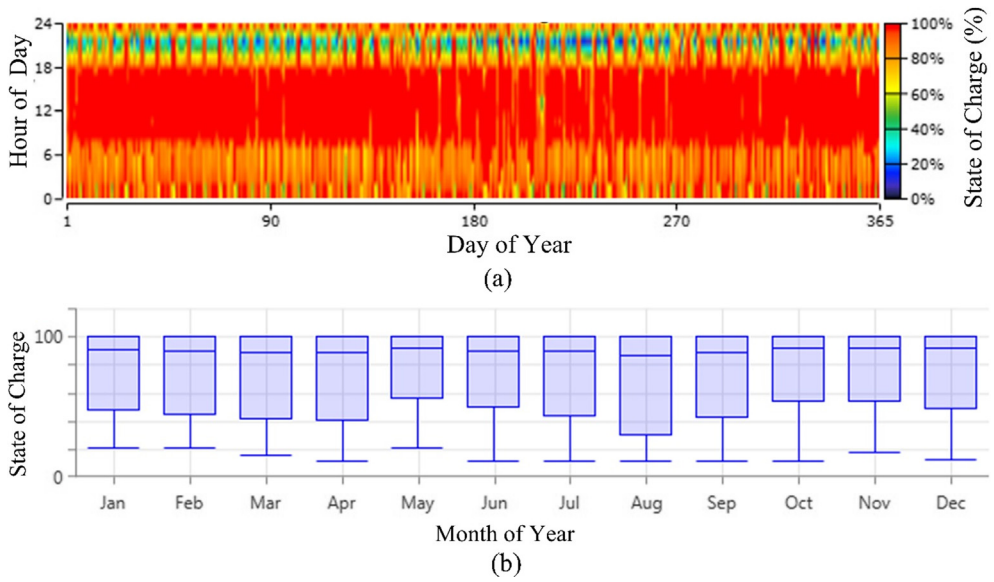


Fig. 15. Lithium-ion battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

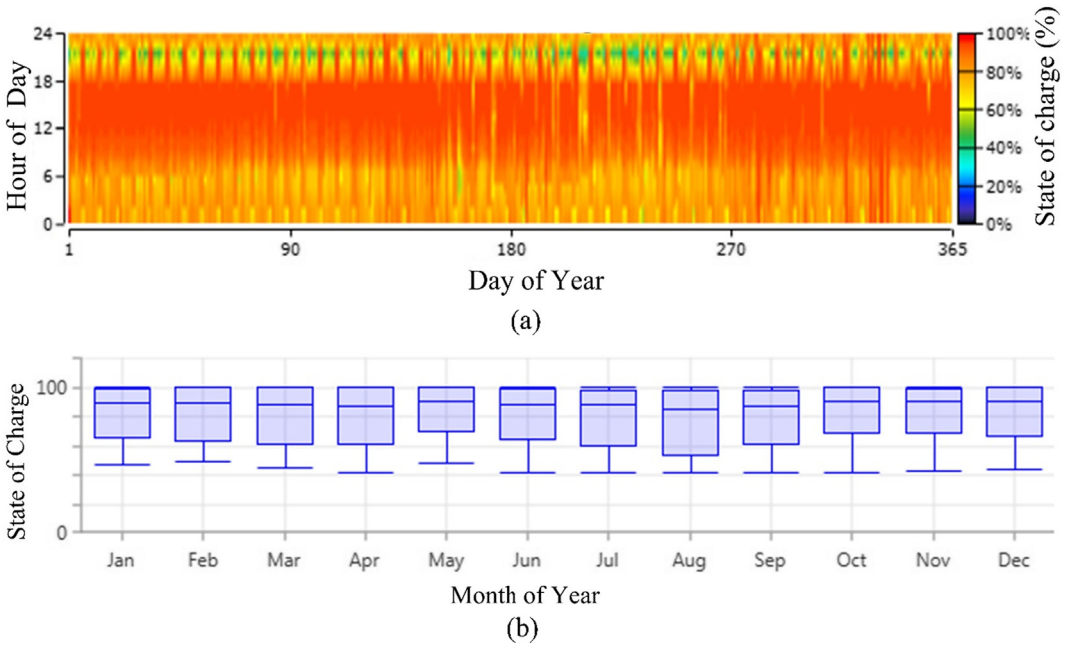


Fig. 16. Lead-acid battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

Table 4
Cost summary for different components used in case-2.

Components		Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV		126,000	0.00	17370.41	0.00	0.00	143370.41
Wind Generator		46,156	0.00	10653.85	0.00	0.00	56809.85
Converter		12,700	5388.28	1158.03	0.00	−2699.24	16547.07
Battery	Li-ion	115,700	48340.13	0.00	0.00	−18994.80	145045.32
	LA	36,332	84761.48	33930.21	0.00	0.00	155023.69
Total System	Li-ion	300,556	53728.41	29140.84	0.00	−21694.04	361731.20
	LA	221,188	90149.76	62238.45	0.00	−2699.24	370876.97

microgrid system having Li-ion battery as a storage medium requires 178 units of batteries, whereas the system having LA battery requires 293 units of batteries for this case scenario. The cycle charging (CC) dispatch strategy has been used in simulation for this scenario. The microgrid supplies continuous power at a cost of 0.12 $/kWh including grid electricity charges. This price is reasonable as against the 0.08 $/kWh [9] price from the unreliable grid. There is a significant reduction in the CO_2 emissions (48,405 kg per annum), as most of the load power demand is fulfilled by the renewable generators (PV, WTG) of the microgrid system.

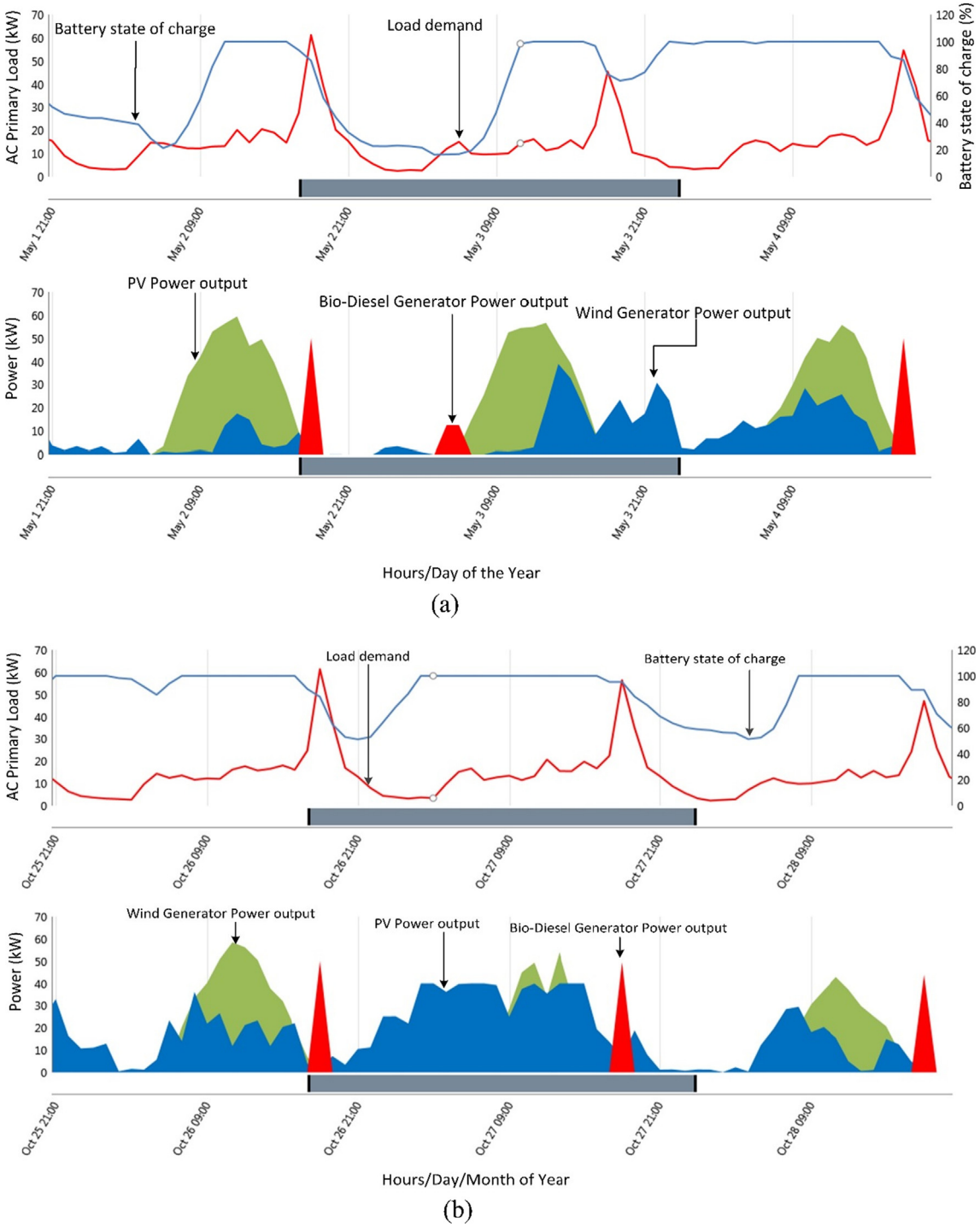


Fig. 17. Energy generation from different sources and battery SOC for the load demand of (a) the second week of May, (b) the fourth week of October.

The load power demand and output power delivered by the generators and grid for the months of July (summer) and December (winter) are shown in Fig. 13(a) and (b), respectively. In these figures, the upper section represents the output waveforms of total primary load, deferral load supplied and electricity exchanged with the grid, whereas the lower section represents the power supplied by PV, WTG and the battery SOC. The obtained result clearly shows that the RER generators fulfil the load power demand and charge the battery to maintain the battery SOC above 70% in most days throughout the year, which leads to an increased lifespan and reduced replacement cost of the battery bank. The load demand is also fulfilled by taking electricity from grid (when available) during the event when the power from RERs generators is not available or insufficient. The excess power generated by the RESs has been sold to the grid on the basis of net metering system, reducing the average cost of electricity. Hence, it can be concluded that the proposed microgrid system satisfies total energy demand with the help of PV, WTG and batteries.

Fig. 14 shows the monthly average energy supplied by the different components of the proposed microgrid system. The PV generator produces most electricity each month due to better availability of solar radiation at the selected location and larger capacity. The WTG also generates a significant amount of power, but only in windy months. The

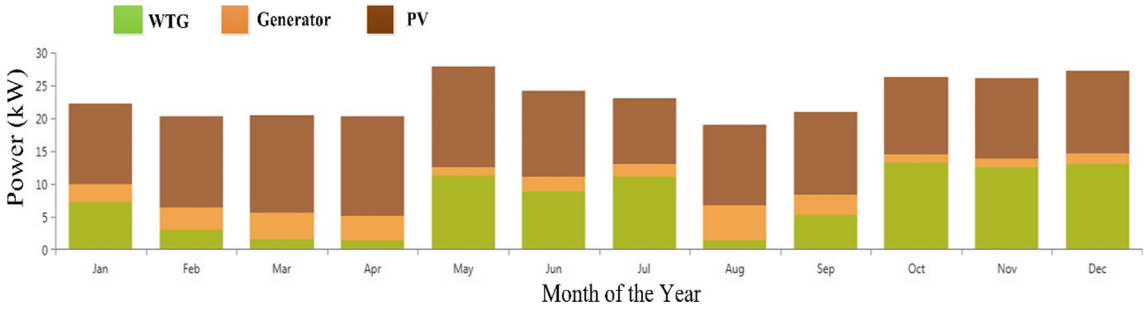


Fig. 18. Average monthly energy generation share from different sources in a year.

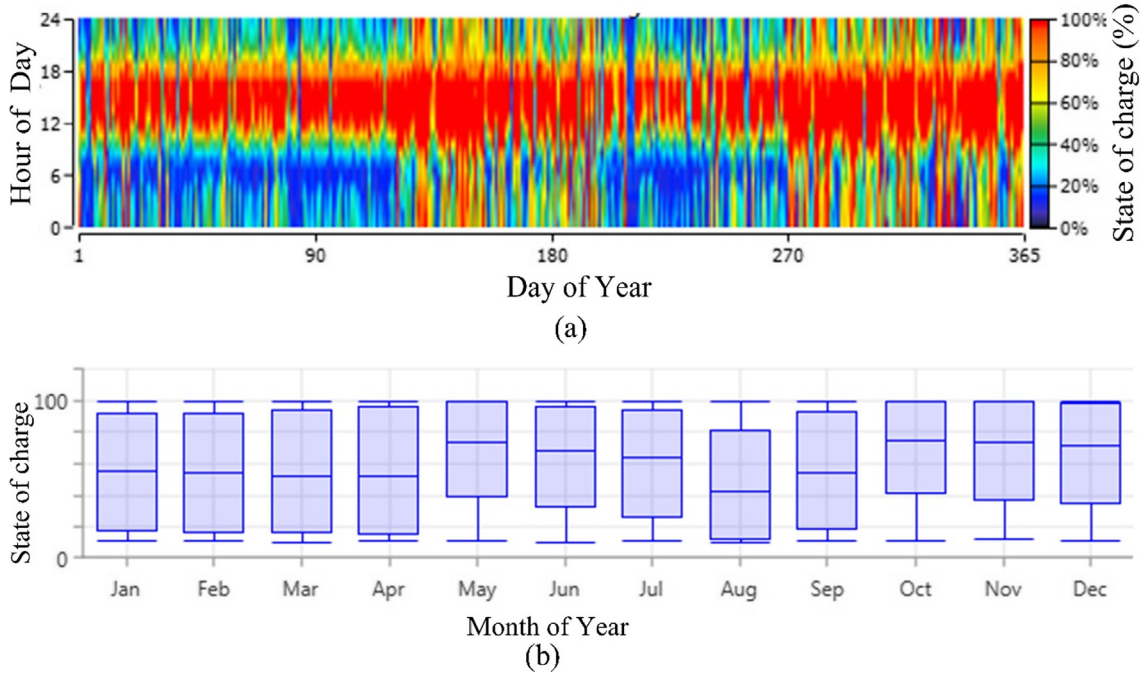


Fig. 19. Lithium-ion battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

electricity is purchased from the grid in the absence of PV and wind generation to fulfil the load demand.

Figs. 15 and 16 give the average SOC of the Li-ion and LA battery unit for each month of the year. As in case-1, the use of the battery bank rises during summer and when additional power is drawn from the batteries. The utilization of both battery types is less than in case-1 due to grid connectivity, which reduces their replacement cost.

The detailed cost analysis of the main components of the optimal microgrid system is presented in Table 4. The net present cost of the whole setup having Li-ion batteries is around $362,000 and for the system having LA batteries is around $371,000. Electricity can be purchased from the grid during higher load demand whenever the grid supply is available, and sold back to the grid during low power demand or excess electricity generation from the PV and WTG at the village. The grid sales bring extra revenue to the system and the overall COE is reduced to approximately 0.12 $/kWh. In this case, also, the type of battery bank has an impact on the COE of the microgrid system. The system with Li-ion batteries provides electricity at 0.122 $/kWh, whereas the system having LA batteries as a storage provides electricity at 0.128 $/kWh. The components that require replacement are the battery bank and converter units. The battery unit has the highest capital cost followed by the PV system as displayed in the Table 4. The operational and replacement cost is dominated by that of the LA battery unit. The set of batteries need replacement according to their lifetime and usages. Li-ion batteries entails 40% of the total cost of the system, whereas LA batteries entail 41.8% of the total cost of the system. In addition, PV contributes 39% and WTG 15% of total lifetime costs in the proposed microgrid system.

4.3. Case-3: PV/WTG/BDG/Battery

The optimal combination of microgrid system components which fulfils the load demand of the residential building are 70 kW PV system, 40 kW WTG, 50 kW BDG, and 49 kW converter with the load following dispatch strategy. The system with Li-ion batteries requires 156 batteries (each 1 kWh) and the system with LA battery type require 273 batteries. The calculated energy cost for an incessant reliable power supply from the microgrid system is to be 0.23 $/kWh. The optimal results of an AC microgrid system having PV, WTG and DG are shown in Figs. 17–20.

The generated power and load demand response by the various components of the microgrid in the month of May and October are shown in Fig. 17(a) and (b), respectively. In these figures, the upper section represents the output waveforms of total primary load, and SOC of the battery, whereas the lower section represents the power generations response from PV, WT and BDG. The results show that the RESs are capable to supply most of the demand and keep the SOC of the battery within limits. The BDG is only operated in the absence of wind and solar energy. In this case also, the proposed microgrid system fulfils the total energy demand with the help of solar, WTG, BDG and

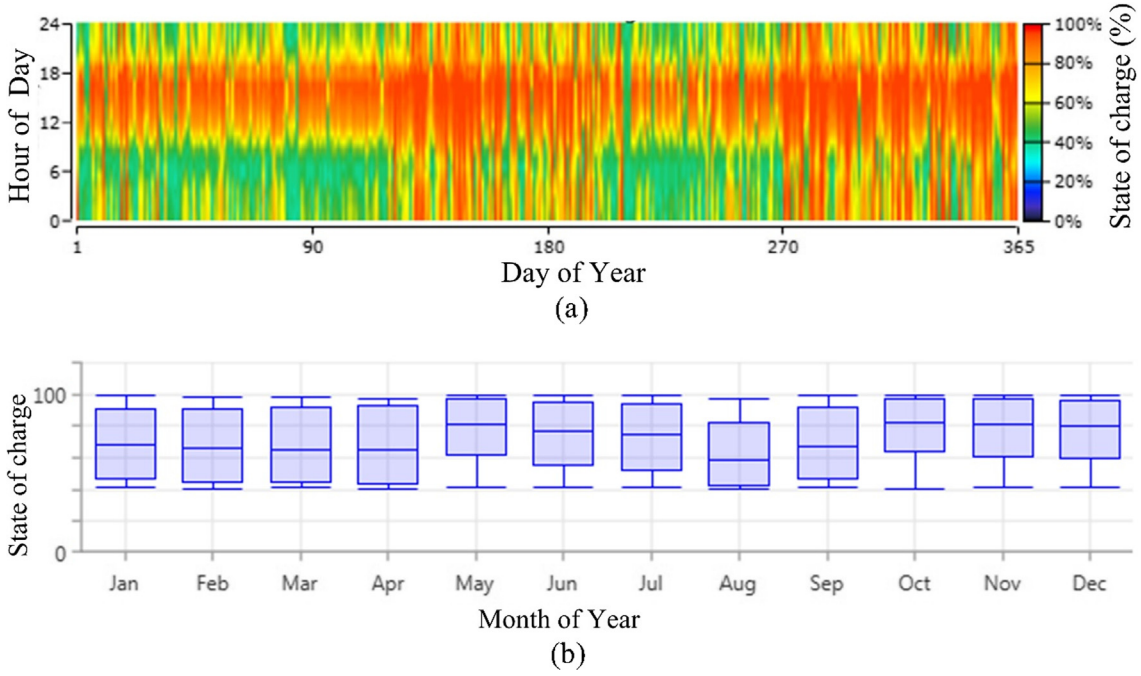


Fig. 20. Lead-acid battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

Table 5
Cost summary for different components used in case-3.

Components		Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV		58,000	0.00	8106.19	0.00	0.00	66906.19
Wind Generator		46,156	0.00	10,653	0.00	0.00	56809.85
Battery	Li-ion	101,400	45482.09	0.00	0.00	−12414.55	134467.55
	LA	33,852	78975.71	31614.15	0.00	0.00	144441.86
Biodiesel Generator	Li-ion	25,000	10742.10	1763.10	49550.61	−5154.05	81901.75
	LA	25,000	10615.93	1738.78	49633.65	−5302.83	81685.19
Converter	Li-ion	7678.96	3257.98	0.00	0.00	−1632.07	9304.86
	LA	7326.71	3108.53	0.00	0.00	−1557.21	8878.04
Total System	Li-ion	239034.96	59482.17	20533.14	49550.61	−19200.67	349390.20
	LA	171134.71	92700.18	52112.97	49633.65	−6860.04	358721.48

batteries.

Fig. 18 demonstrates the monthly average energy balance for one year. The selected location in this case has better availability of natural resources, therefore a good amount of solar and wind power is generated. If the generated power by the PV panels and WTG are less, then BDG comes into operation to fulfil the load demands.

Figs. 19 and 20 give the average SOC of the Li-ion and LA battery unit for each month of the year. From the obtained results, it can be inferred that the SOC remains within the specified limits for Li-ion (10–100%) and LA (40–100%) batteries, respectively. It can be observed from the figures that the usage of the battery bank is highest in the months of January, March, April, August and September.

Table 5 shows the detailed cost analysis of the optimal microgrid system. The net present cost of the whole setup having Li-ion batteries is around $350,000 and for the system having LA batteries is around $359,000. In this case also, the system with Li-ion batteries has a lower cost of electricity at 0.23 $/kWh, whereas the system with LA batteries costs 0.237 $/kWh. The lifetime of BDG, hinge on the operating hours instead of life in years. Here the operating hours of BDG is low, hence no replacement is required. The components that require replacement are the battery bank and the converter units. Again, battery unit has the highest capital cost followed by the PV system. The O&M cost is also dominated by the battery units.

4.4. Case-4: PV/WTG/BDG/Battery

The optimal combination of microgrid system components to fulfil the load demand of the IT building is 1000 kW PV, 290 kW WTG, 360 kW BDG, and 500 kW converter with load following dispatch strategy. This system with Li-ion batteries requires 568 kWh storage, whereas the system with LA batteries requires 1031 units. The calculated COE for continuous supply of power is between 0.138 and 0.142 $/kWh. The optimal results of an AC microgrid system having PV, WTG and BDG are shown in Figs. 21–24.

The response of the designed microgrid system in the form of power generated by the various generators, load power demand and battery SOC in the month of May and November are shown in Fig. 21(a) and (b), respectively. As before, the upper section represents the output waveforms of total primary load and deferral load, while the lower section shows the power supplied by PV, WTG, BDG and the SOC of the battery. The proposed microgrid system fulfils total energy demand with the help of PV, WTG, batteries and the BDG is only operated in the absence of wind and solar energy.

Fig. 22 shows the monthly average energy supplied for each month from different generator types. The annual average electricity production is dominated by the PV due to higher rating and good solar availability. When the power generated by PV panels and WTG are not sufficient or less, then the load demand is fulfilled by BDG.

Figs. 23 and 24 demonstrate the average SOC of the Li-ion and LA

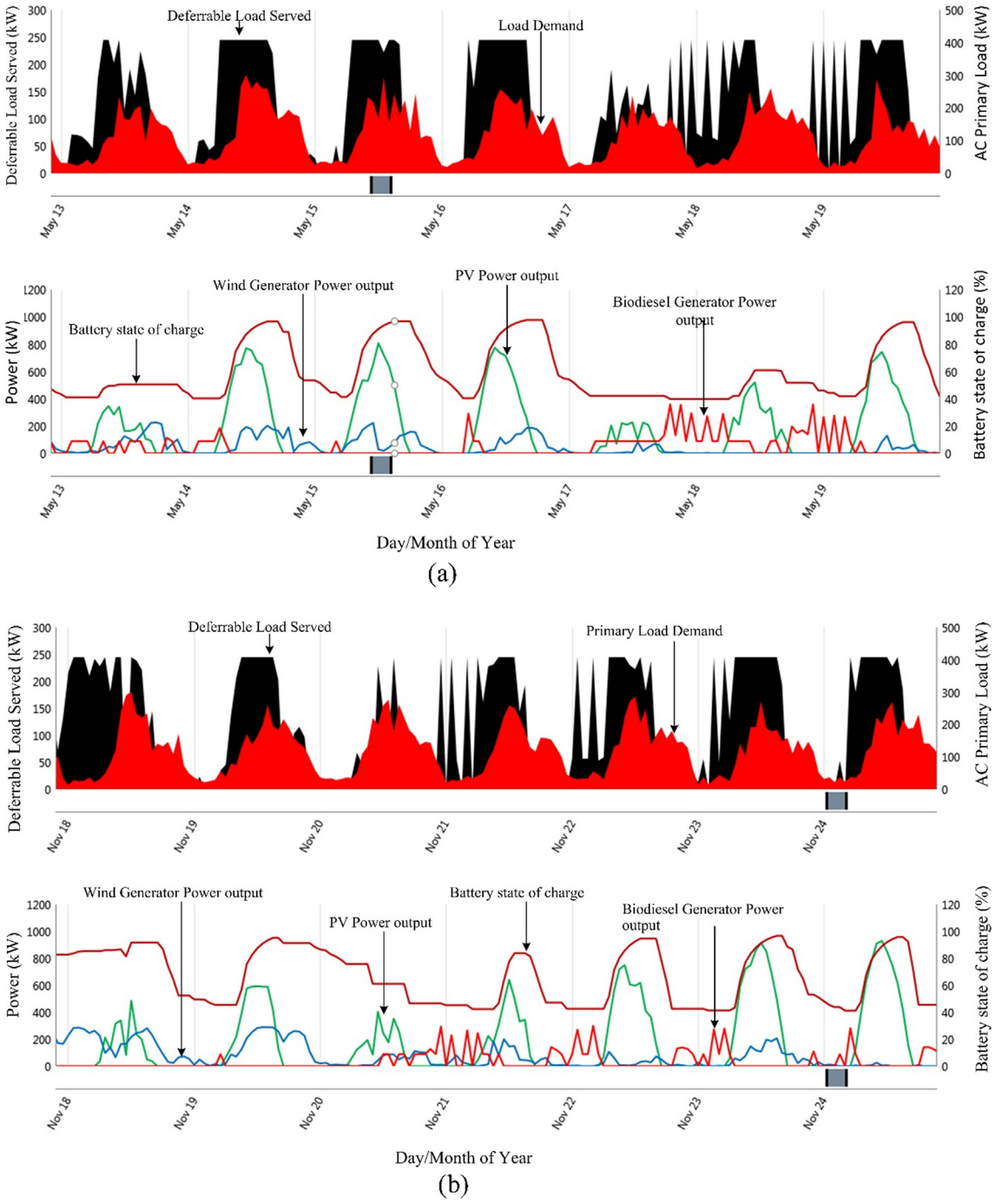


Fig. 21. Energy generation from different sources and battery SOC for the load demand of (a) the second week of May, (b) the third week of November.

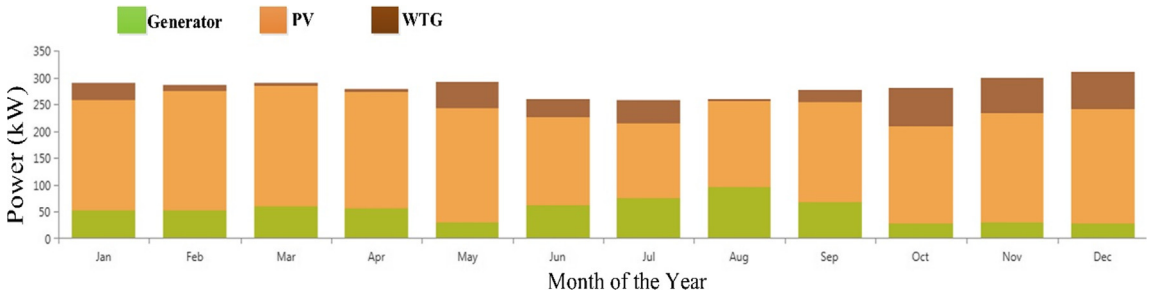


Fig. 22. Monthly average share of energy generation from different sources for a year.

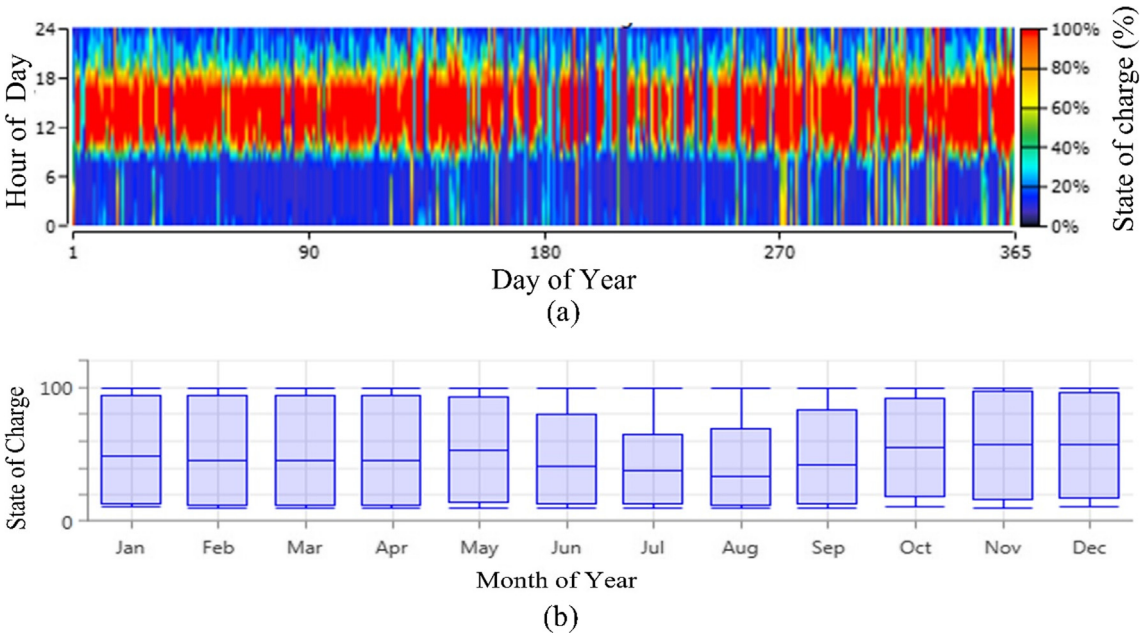


Fig. 23. Lithium-ion battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

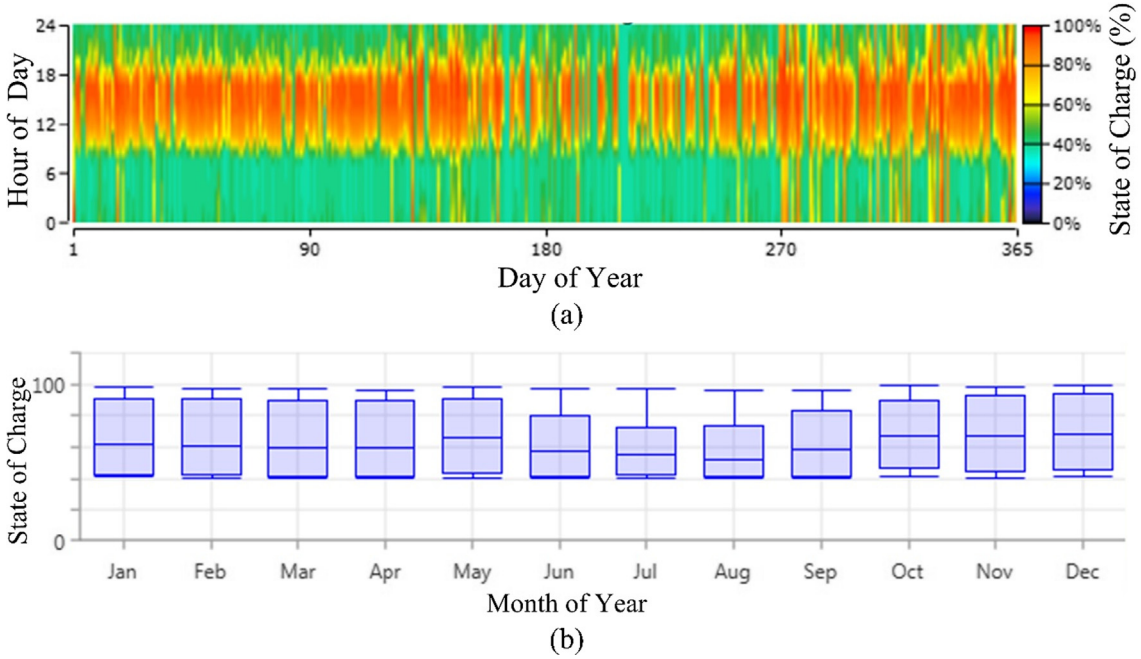


Fig. 24. Lead-acid battery SOC (a) Hours of the day vs. day of the year, (b) SOC vs. month of the year.

Table 6
Cost summary for different components used in case-4.

Components		Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV		840,000	0.00	115802.75	0.00	0.00	955802.75
Wind Generator		334,631	0.00	77240.43	0.00	0.00	411871.43
Battery	Li-ion	369,200	167407.59	0.00	0.00	−42575.56	494032.03
	LA	127,844	298256.27	119392.64	0.00	0.00	545492.91
Biodiesel Generator	Li-ion	138,600	312038.62	608.49	919406.8	−14552.15	1356101.7
	LA	138,600	312325.97	609.53	920872.8	−14198.66	1358209.7
Converter	Li-ion	65029.52	27590.32	0.00	0.00	−13821.26	78798.58
	LA	63097.14	26770.46	0.00	0.00	−13410.55	76457.05
Total System	Li-ion	1747460.5	507036.53	193651.67	919406.8	−70948.98	3296606.5
	LA	1504172.1	637352.70	313045.35	920872.8	−27609.21	3347833.8

Table 7
Comparative analysis of microgrid configurations based on different performance indicators.

Components		PV (kW)	WTG (kW)	Diesel/ Biodiesel Generator (kW)	Batteries (Units)	Gen. Running (Hrs.)	Inverter (Units)	Total NPC ($)	Operating Cost ($)	COE ($/kWh)	Renewable Fraction (%)	Dispatch strategy	Grid connected
Case-1	Li-ion	300	20	105	786	360	103	1,123,082	19044.10	0.225	93.5	CD	No
	LA	300	20	105	1336	364	102	1,288,555	56372.02	0.258	93.5	CD	
Case-2	Li-ion	150	40	–	178	–	100	361,731	5282.71	0.122	70.5	CC	yes
	LA	150	40	–	293	–	100	370,877	12,926	0.128	71.0	CC	
Case-3	Li-ion	70	40	50	156	1015	49	349,390	9530	0.230	82.0	LF	No
	LA	70	40	50	273	1010	48	358,721	16,199	0.237	82.0	LF	
Case-4	Li-ion	1000	290	360	568	3504	512	3,296,606	133,775	0.138	77.1	LF	No
	LA	1000	290	360	1031	3505	500	3,347,834	159,207	0.142	77.1	LF	

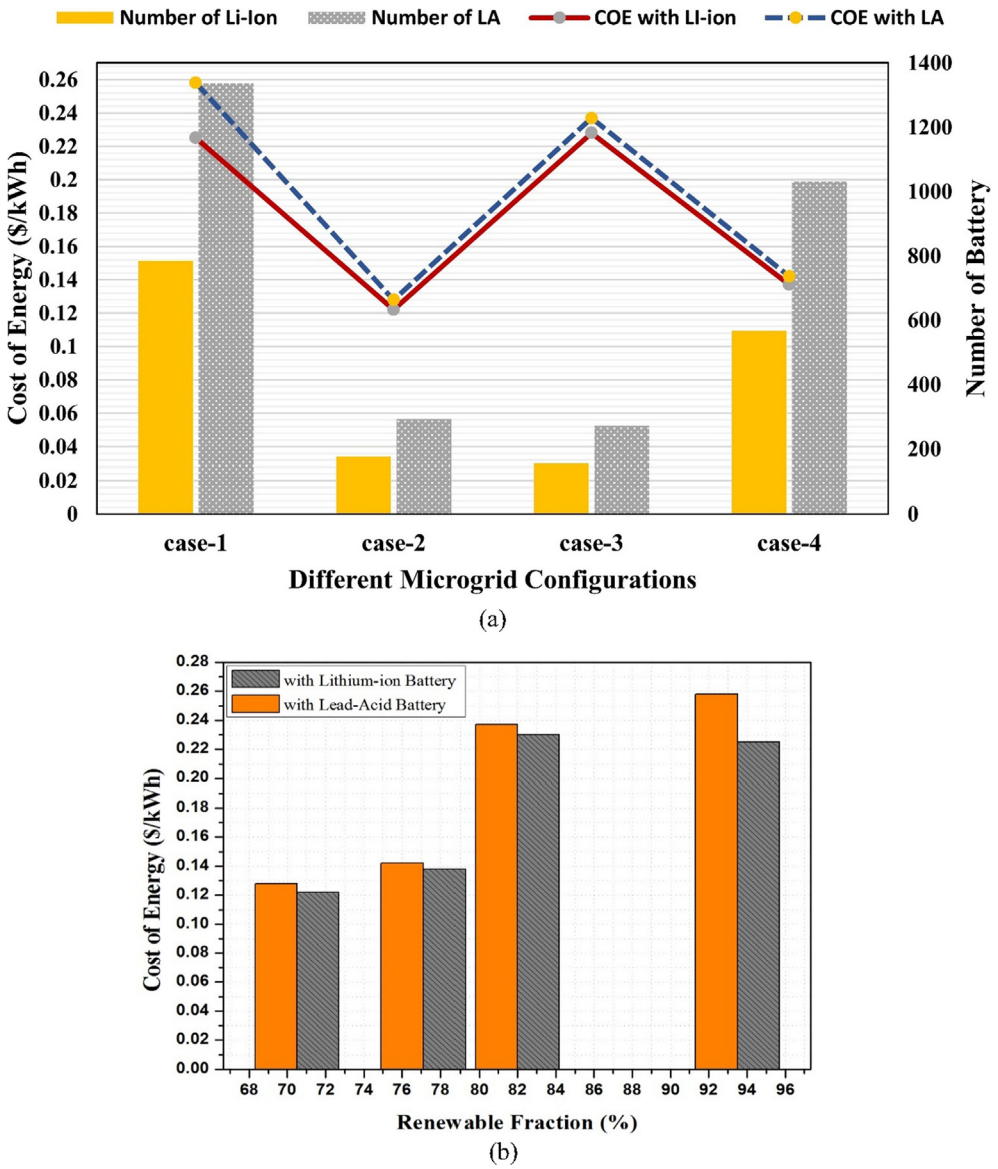


Fig. 25. Comparative analysis of various microgrid configurations in the presence of different type of batteries (a) Number of battery/Cost of energy vs different microgrid configuration. (b) Cost of energy vs renewable fraction (%).

battery unit for each month of the year. The SOC is retained within the prescribed limits for each battery type and maximum usage is in the months of May to August.

Table 6 shows the detailed analysis of the various cost components of the optimal microgrid system. The net present cost of the whole setup having Li-ion batteries is around $3.30 million and for the system, having LA batteries is more expensive at around $3.35 million. Per-unit COE of the system having Li-ion batteries (0.138 $/kWh) is also lower than the system with LA batteries (0.142 $/kWh) as a storage medium.

The detailed comparative analysis of the simulation results obtained for the four case studies is presented in Table 7. The main focus of this study is to compare the impact of battery type between Li-ion and LA

Table 8
Key results of the Li-ion and LA battery in various microgrid configurations.

Parameters	Case-1		Case-2		Case-3		Case-4	
	Li-ion	LA	Li-ion	LA	Li-ion	LA	Li-ion	LA
Energy In (kWh/yr)	179,775	201,646	43,287	49,307	41,617	46,252	153,563	175,707
Energy out (kWh/yr)	162,263	161,899	38,964	39,490	37,507	37,276	138,648	141,032
Storage depletion (kWh/yr)	491	651	6.53	49.9	54.8	61.9	465	521
Losses (kWh/yr)	18,003	40,398	4329	9867	4164	9312	15,380	35,196
Annual Throughput (kWh/yr)	171,040	181,008	41,072	44,151	39,536	41,675	156,148	157,678
Expected life (yr)	14.7	4	13.9	4	12.6	4	12.4	4

Table 9
Greenhouse gases and emissions reductions.

Quantity	Case-1	Case-2	Case-3	Case-4
Carbon dioxide (kg/yr)	18,496	48,405	−207	393,047
Carbon monoxide (kg/yr)	120	0	132	1881
Unburned Hydrocarbons (kg/yr)	5.09	0	5.81	108
Particulate Matter (kg/yr)	0.482	0	0.791	108
Sulphur Dioxide (kg/yr)	45.3	210	0	960
Nitrogen Oxides (kg/yr)	9.61	103	124	2151

batteries to find out which is most suitable economically and technically. Therefore, different microgrid configurations have been taken to justify the comparative findings. In each case study, the composition and the rating of the components (WTG, PV, converter, load, etc.) used are exactly same (as displayed in Table 7) except the types of batteries.

The per-unit COE and TNPC of the system are the two important factors to evaluate the economic feasibility of any microgrid. The COE for each microgrid system mainly depends upon the renewable fraction, dispatch strategies employed, types of the batteries used and grid connection. The load following, cycle charging, and combined dispatch strategies have been used according to the operating conditions and to reduce the operational cost of the microgrid system. The strategy, in which the DG automatically triggers once the battery touches a certain minimum voltage level is determined by its maximum discharge depth called cyclic charge. The phenomenon in which the DG only supplies the load, while the battery is recharged only from the available renewable resources is termed load following strategy. The strategy in which the excess energy from renewable sources is always used to charge the battery whenever required, and run the DG set if the critical load is lower than the net load is called combined dispatch strategy. The detailed description about these strategies is given in Refs. [9,19].

Fig. 25(a) shows the graphical representation of number of batteries required in the different configurations of microgrid. It can be seen from this bar-chart that the number of Li-ion batteries in each case is less than the number of LA batteries. The reduced number of batteries affects the per-unit COE. The COE of the microgrid systems with Li-ion batteries is also lower relative to LA batteries in all four cases, as displayed in the Fig. 25(a). In addition, the DG operating hours have been reduced due to battery bank, as the stored power is fed to the load when required and allowing the DG to operate at an optimum loading point. This shows that the battery improves the life span of the generator besides reducing greenhouse emissions.

The fraction of the energy supplied to the load that is generated from renewable energy resources (WTG, PV) is termed the renewable fraction. Fig. 25(b) shows the variation of the COE with respect to renewable fraction. The microgrid systems with higher renewable fraction in total electricity generation have higher COE or vice-versa. In addition, a system having higher renewable fraction requires a higher number of batteries to store the energy generated by using RES or vice-versa. Because of this, the overall cost of the system as well as the COE have also increased as shown in Table 7, but in each case, the microgrid system with Li-ion batteries has lower COE as compared to the system with LA batteries.

The whole analysis revealed that the COE and TNPC cost of every microgrid system in the proposed study are highly dependent on the type and number of batteries utilized to store the energy. The requirement of the batteries reduces up to 45% when Li-ion batteries are used as a storage medium as shown in Table 7. Similarly, Li-ion batteries have lower lifetime costs than LA batteries when used in intermittent microgrid systems, resulting in an average 6% reduction in the COE, ranging between 2.8% and 13%. Overall, the study reveals that the use of Li-ion batteries for the application of storage in all four configurations of microgrid system proves to be economical.

The key results of the Li-ion and LA battery simulation used in the four microgrid configurations are shown in Table 8. It can be clearly seen from the results that Li-ion battery has out-performed the LA battery in terms of overall losses (kWh/yr), storage depletion rate (kWh/yr) and expected lifetime (yr). The analysis (in Table 8) revealed that Li-ion batteries have longer life, low losses, and extended cycle life with lesser storage depletion rate as compared to LA batteries. One of the main reasons why the COE is lower for microgrids with Li-ion batteries is that the overall losses reduce from around 20% of generated energy to around 10%.

Table 9 shows the greenhouse (GHG) emissions statistics in the different configurations of the proposed microgrid system. The RES system can reduce GHG emission by a significant amount, and is thus very environment friendly. In addition, biodiesel is an environment friendly fuel as compared to diesel. Therefore, there is significant reduction of the GHG emissions in the microgrid system having BDG (Case-3 and 4) as compared to the microgrid system consisting of DG (Case-1) as displayed in Table 9. The microgrid system with BDG can reduce GHG emission by a significant amount, and therefore biodiesel can be promising as an alternative fuel for the diesel engines.

5. Conclusion

This study evaluates the techno-economic analysis of lithium-ion and lead-acid batteries, and assess their ability to manage the load demands in various configurations of the microgrid. The performances of both the batteries have been studied in the proposed microgrid system using realistic load profiles, real resources data and real prices of components. A detailed analysis of microgrid systems in isolated as well as grid-connected mode for rural and urban scenarios has been done. The key conclusions of this study are summarized as follows:

- Batteries help in providing reliable power supply and aid the generators (PV, WTG, DG/BDG) to fulfil the total energy demand in the proposed microgrid systems.
- Both types of battery effectively store excess power during low power demand and feed the stored power during high demand periods.
- The utilization of both types of battery in a grid connected microgrid system is lower than for off-grid microgrids.
- A system having higher renewable fraction requires higher number of batteries bank or vice-versa.

- The battery bank reduces the diesel generator operating hours, which results in longer generator life and reduced greenhouse emissions.
- The grid connected rural microgrid (case-2), is economical in terms COE because the grid sales bring extra revenue to the system and the overall COE reduces.
- The microgrid system with Li-ion batteries, as a storage medium require up to 45% lesser batteries, have lower net present cost and reduced COE as compared to LA batteries.
- Li-ion batteries have low losses, and extended cycle life with lower storage depletion rate as compared to LA batteries in all of the systems studied.

Apart from the costs, there are additional advantages of Li-ion batteries, which will apply in all cases. These include reduced maintenance, more flexibility in SOC, fast charging rate, smaller volume and longer cycle life. Considering various factors obtained from the studies carried out, it can be concluded that lithium-ion batteries should be recommended as an alternative viable solution over lead-acid batteries in various applications of future electric power systems.

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